

Where Win Probability Comes From, and Where It Breaks

Spatial Control, Bomb Geometry, and an Out-of-Time Reality Check for Live Win Probability in Counter-Strike 2

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Abstract

Live win-probability models are widely deployed in esports broadcasting, yet the published Counter-Strike literature evaluates them almost exclusively on economy and combat state, pooled across all game situations, and validated only by cross-validation. We build a per-second win-probability model for Counter-Strike 2 on the map *de_inferno* from 220 professional matches (476,595 per-second snapshots) and use it to examine three questions that pooled AUC obscures: where the predictive signal is concentrated, which spatial representation is most predictive, and whether the model’s performance survives temporal distribution shift.

We report four findings. First, the conventional AUC of approximately 0.85 is attributable predominantly to lopsided game states. Restricted to even rounds, defined as equal players alive and near-equal equipment, every model attains an AUC of approximately 0.58. We term this conditional measure *contested-AUC* and argue that it, rather than pooled AUC, is the appropriate criterion for a live model. Second, we adapt Voronoi pitch control from association football and compare three map-control representations of increasing physical fidelity. Fidelity and predictiveness diverge: an instantaneous line-of-sight, field-of-view, and smoke model is the most realistic representation and the least predictive, and it regains parity with proximity-based control only when augmented with temporal memory. Predictive value derives from the temporal stability of the representation rather than its instantaneous accuracy. Third, a physically motivated *defuse-race* feature, defined as the fuse time remaining minus the nearest counter-terrorist’s path time and defuse duration, is the strongest single addition beyond economy and reduces post-plant log-loss by 7–8%. Fourth, across nine architectures (logistic regression, random forest, XGBoost, LightGBM, CatBoost, a causal temporal convolutional network, a causal Transformer, a player-graph attention network, and an ensemble), evaluated with match-level block-bootstrap confidence intervals, the models are statistically indistinguishable, both in cross-validation and out-of-time.

We then evaluate the model once on a temporally held-out set of 27 unseen 2026 matches. The features derived from the match itself transfer with negligible degradation (0.8493 $\rightarrow$ 0.8501), whereas the best cross-validated configuration degrades most (0.8519 $\rightarrow$ 0.8236; expected calibration error 0.016 $\rightarrow$ 0.071) because its player-skill pillar depends on an external, per-era statistics database that is incomplete for the deployment period. This failure mode is invisible to cross-validation by construction. Reconstructing the prior under three data regimes, including a leak-free lagged prior and a leaky same-era prior at full coverage, restores calibration but does not enable the skill pillar to exceed the skill-free model out-of-time under any regime. Nor does re-encoding it: across four encodings that successively remove a player-count confound, adjust for opponent calibre and condition on situation, cross-validation prefers every one of them and the out-of-time test prefers none. The result

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is a controlled negative finding: a correctly constructed, coverage-complete and correctly encoded external skill prior does not improve out-of-time performance in this setting.

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1 Introduction

Win probability converts a complex game state into a single interpretable quantity and is a standard component of live sports analytics. In esports the underlying state is unusually detailed: each player’s position, velocity, view angle, health, equipment, and utility are recorded at 64 Hz. Nonetheless, published Counter-Strike win-probability models remain dominated by economy features such as equipment value, players alive, and bomb status. The spatial component of the game, which is central to expert reasoning, is largely unmodelled.

We identify three methodological conventions in this literature that obscure how predictive these models actually are. First, evaluation is pooled: a single AUC computed over all snapshots aggregates decided situations (for example, five players against one) with genuinely uncertain ones (five against five on equal economy), and a model that captures only headcount and equipment scores well because most snapshots are easily classified. Second, physical fidelity is assumed to imply predictiveness: representations that model line of sight, field of view, and smoke occlusion are presumed superior because they more faithfully describe the game. Third, cross-validation is treated as the final validation step, even though random or grouped folds are drawn from the same period as the training data and cannot expose failures that arise under temporal distribution shift.

1.1 Contributions

- **Contested-AUC** (Sect. 7.3). We introduce a conditional evaluation measure restricted to snapshots with equal players alive and near-equal equipment. Under this measure the economy baseline falls from 0.846 to 0.580, demonstrating that pooled AUC substantially overstates model performance in the situations that determine round outcomes.
- **A map-control ablation** (Sect. 7.6). We compare three representations of increasing physical fidelity and find that the most faithful is the least predictive, and that adding temporal memory restores its predictive value. Predictiveness follows from temporal stability rather than instantaneous fidelity.
- **Defuse-race geometry** (Sect. 7.7). A single physically derived feature is the strongest addition beyond economy, reducing post-plant log-loss by 7–8% without degrading calibration.
- **An out-of-time evaluation of nine architectures** (Sects. 7.10, 7.15). Sequence and graph neural models are statistically indistinguishable from a calibrated logistic regression, in cross-validation and out-of-time. On a temporally held-out 2026 set, the player-skill pillar degrades sharply through a dependence on an external, per-era database, while the features derived directly from the match transfer intact.

The final contribution is the most general: it concerns evaluation protocol for models that incorporate externally sourced features, and is not specific to Counter-Strike.

2 Related Work

Counter-Strike win probability. Xenopoulos et al. introduce a per-round win-probability model and action-valuation framework for Counter-Strike built on economy and combat state [15], together with the ESTA trajectory-and-action dataset [16]; this establishes the convention we adopt as our Model A and extend. Such models predict at round granularity from a frozen post-buy snapshot. We instead predict at every second of the round, as a live overlay requires, which changes the evaluation problem: most per-second snapshots are near-deterministic, motivating the contested-AUC measure of Sect. 7.3. All demos are parsed with the `awpy` library [17].

Spatial control in team sports. Voronoi-based dominant-region and pitch-control models are well established in team-sport tracking analytics, originating with the dominant-region formulation of Taki and Hasegawa [12] and developed for football by Fernández and Bornn [4] and Spearman [11]. Their transfer to a tactical shooter is not obvious. Football pitch control is governed by reachability under a motion model; in Counter-Strike, control is mediated by sight-lines, weapon range, utility that blocks vision, and, as we show, memory of previously cleared space. A player need not be able to reach a corridor to control it; he needs to be able to fire down it. This asymmetry makes the direct Voronoi transfer a non-trivial baseline rather than an obviously correct one.

Deep models on tracking data. Graph attention networks over player nodes [14], temporal convolutional networks over state sequences [1], and Transformer encoders [13] are standard architectures for tracking-derived prediction. Our result here is negative but carefully quantified: at 220 matches, none beats a calibrated linear model with engineered features (Sect. 7.10).

Calibration and evaluation. We adopt proper scoring rules (log-loss and Brier), the Murphy decomposition of the Brier score [7], calibration slope and intercept, and binning-free calibration estimates. Calibration is frequently under-reported in win-probability work despite being essential for a live product, and modern high-capacity models are known to be poorly calibrated by default [5]. Confidence intervals on all metrics use a match-level block bootstrap, and AUC differences are corroborated with the test of DeLong et al. [3].

3 Data and Pipeline

3.1 Source and scope

220 professional *de_inferno* matches (2024–2025, Tier-1 events), parsed with `awpy` v2.0.2. We restrict to a single map so that spatial features are directly comparable across matches: a nav-mesh area index, a zone label and a control fraction only mean the same thing across two matches if the geometry is identical. Generalising across maps is left to future work (Sect. 9).

3.2 Match selection (inclusion criteria)

The selection rule is fixed in advance and applied uniformly, so the corpus is reproducible rather than curated. We include every *de_inferno* map played at a defined set of **Tier-1 events** from 2024–2025: the four Valve Majors of the period (PGL Major Copenhagen 2024, Perfect World Shanghai Major 2024, BLAST Austin Major 2025, StarLadder Budapest Major 2025) together

with the top-tier IEM, ESL Pro League, BLAST Premier, and PGL circuit stops (IEM Katowice, Cologne, Dallas, Chengdu; ESL Pro League Seasons 20–21; BLAST Premier Spring/Fall/World Finals; PGL Cluj-Napoca 2025). We take *every* Inferno map from these events, including both mirror demos of a match where both were recorded, and apply no filtering on team strength, round count, or outcome. A demo is dropped only if it fails the mechanical validation below, never for its content. Under this rule 264 Inferno demos are collected and 44 are removed by validation, leaving the 220 used throughout. The full list, with event, stage, and result for each demo, is released with the code so the corpus can be reconstructed exactly.

Parsed demos are written to columnar *channels*, ticks, rounds, bomb events, kills, smokes, and the feature assembler joins them per snapshot. Keeping the channels separate rather than materialising one wide table early is what later made the out-of-time holdout possible to rebuild from scratch without contaminating training (Sect. 7.15).

3.3 Snapshots and labels

Each round is sampled *once per second* from freeze-end to round end. Every snapshot is one training row, labelled with the round’s eventual winner (`ct_won`). This yields **476,595 snapshots** over $\approx 4,900$ rounds, base rate $P(\text{CT win}) = 0.445$. Snapshots within a round share a label; the probability moves because the *state* moves.

3.4 Three parsing pitfalls worth reporting

These are unglamorous and they cost real time. We report them because each is silent, none produces an error, and none is visible in an aggregate metric.

1. **Tickrate.** The demo header’s tickrate is unreliable on a minority of files. We force 64. A mis-read tickrate silently rescales *every time-based feature*, including the defuse-race margin of Eq. 1, which is measured in seconds and would be wrong by a constant factor on the affected demos.
2. **The halftime swap.** Trusting the per-tick side field inverts every label in the second half. We recover side assignment via team-clan mode instead. This bug is invisible in aggregate metrics (it just quietly caps your AUC) and is the kind of error that a model with enough capacity will partially *absorb*, which makes it worse, not better.
3. **Clinch-trimmed rounds.** Demos that end mid-round produce rounds with no resolution. Dropped.

In total, 44 of 264 parsed demos fail validation and are excluded, leaving the 220 used throughout.

3.5 The out-of-time test set

A separate bundle of **27 unseen 2026 matches** (55,271 snapshots; IEM Kraków, the Cologne Major, Atlanta, BLAST Rivals), zero overlap with training, is held out and touched **exactly once**, at the very end (Sect. 7.15). Its base rate is 0.512: 2026 Inferno is measurably more CT-sided than 2024–25 Inferno. That is itself a distribution shift, and we deliberately do *not* correct for it, because correcting for it would conceal exactly the kind of drift we want the holdout to expose.

The raw 2026 channels live in a physically separate directory tree and are assembled through a separate code path, so that no glob over the training directories can sweep them in. This sounds paranoid; it took one near-miss to justify.

4 Feature Pillars

We organise 115 features into four pillars. Feature-set names used throughout: **A** = economy and combat state; **E** = A + map control + tactical/bomb; **EB2** = E + defuse-race; **F** = A + firepower; **EF** / **EFB2** = the corresponding combinations with firepower, carrying the summed, situationally-gated encoding of Sect. 4.4.

4.1 Pillar 1, Economy and combat state (17 features)

The literature baseline. Despite the conventional name “economy”, this pillar covers the full basic game state, not only money: equipment value, health, armour, players alive, defuse kits, score, time elapsed, and the bomb-planted flag. We keep the name “economy” (abbreviated **A**) for continuity with prior work, but the reader should note that combat-state variables such as players alive, team health, and round time live here, and they are what most of the pillar’s predictive power comes from. All quantities are aggregated over *alive* players only, since a dead player’s rifle is not a resource. This pillar alone reaches AUC 0.8465 and is the yardstick every other pillar is measured against.

4.2 Pillar 2, Map control, three ways

We adapt Voronoi pitch control to the *de_inferno* navigation mesh (3,060 areas). For each snapshot we compute, for every nav area, which team controls it, then aggregate to area-weighted control fractions (overall and per macro-zone (A site, B site, banana, mid, CT spawn)) plus a rolling trend and a volatility term. The three representations differ *only* in how an area is assigned, which is what makes the comparison clean.

- (a) **Voronoi (proximity), 9 features.** Each nav area is assigned to the nearest *living* player’s team via a KD-tree. Crude: it ignores walls, vision and smokes, and will happily award a team the inside of a room it cannot see into. Every area is assigned to somebody.
- (b) **Grey (instantaneous LOS + FOV + smoke), 5 features.** An area is controlled only if a living player is within weapon range *and* has line-of-sight to it (a precomputed 3060×3060 nav visibility matrix, ray-cast against the map mesh) *and* is facing it (a field-of-view test against the player’s yaw) *and* it is not occluded by an active smoke. Areas nobody satisfies are “grey”. The result is a four-state surface (CT / T / contested / grey). This is by far the most physically faithful of the three.
- (c) **Territory (grey + memory and decay), 5 features.** The grey model, but *cleared space stays yours*. An area marked by a team remains theirs for a decay window of 15 s without needing re-checking; the FOV test then only gates *re-acquiring* neglected space, not *holding* space already taken; and a player always holds his immediate surroundings regardless of where he is looking. Stateful within a round. The 15 s window is motivated by the map’s geometry: an opponent cannot teleport across cleared ground, so a corridor a team has just held remains effectively theirs for the time it would take an enemy to physically traverse and re-contest it. A short memory therefore encodes control that persists between glances, whereas the instantaneous grey model discards it the moment a player looks away. We fixed the window at 15 s, near the lower end of the range over which the territory features are most predictive, so that the memory reflects realistic traversal times on *de_inferno* rather than an arbitrary long horizon.

Figure 1 makes the difference concrete, and is worth studying before reading the ablation in Sect. 7.6. Voronoi (top row) partitions the *entire* map, CT 67 %, then 64 %, then 44 % as the T side takes space. The grey model (middle row) says something completely different about the same three moments: instantaneous CT control is only 15 %, 20 %, 11 %, and most of the

map is grey, genuinely held by nobody. The grey surface is unmistakably the truer picture of Counter-Strike, yet Sect. 7.6 shows it is the worse predictor. The territory model (bottom row) restores stability: with a short memory of cleared space, CT control recovers to 24 %, 44 %, 27 %, tracking the Voronoi trend without Voronoi’s physical implausibility.



Figure 1: The same round (FaZe vs G2, round 5) at three moments, under the three control models. **Top:** Voronoi proximity control partitions the whole map. **Middle:** the instantaneous LOS+FOV+smoke “grey” model, which additionally requires line-of-sight, facing, and no smoke occlusion; most of the map is grey, held by nobody. **Bottom:** the territory model (grey plus a 15s memory of cleared space), which recovers a stable, Voronoi-like reading without awarding unseen ground. Blue = CT, orange = T, purple = contested, grey = neutral; short lines show facing; white discs are smokes. The grey row is the most faithful description of what players can contest at an instant, but the territory row is what predicts the round (Table 4).

4.3 Pillar 3, Tactical readiness and bomb geometry (42 features)

Utility inventories (per-type grenade counts per side), AWP presence and the zone the AWP is in, per-zone player counts for both sides, positional (Shannon) entropy as a measure of how spread out a side is, and bomb geometry: plant site, plant coordinates, and the nearest CT's straight-line *and* nav-mesh Dijkstra path distance to the bomb.

The straight-line/path distinction matters more than it sounds. On Inferno a CT can be 10 metres from the bomb through a wall and 30 seconds away through the map. Figure 2 shows the consequence: the SHAP dependence for `min_ct_dist_to_bomb` is smooth and monotone-ish, i.e. the model treats raw proximity as a soft signal, whereas the *path-derived* defuse margin below produces a hard step. The physics is in the path, not the distance.

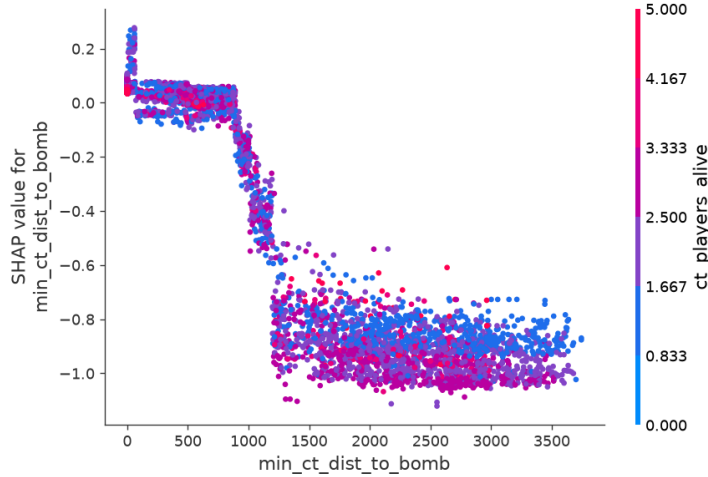


Figure 2: SHAP dependence for the nearest CT's straight-line distance to the bomb. A soft, graded effect, compare with the sharp feasibility boundary the path-derived defuse margin produces (Fig. 9).

The defuse race. Post-plant, the round stops being a fight and becomes a race against the fuse. We track the bomb's live state (*carried* / *dropped* / *planted*) from the event stream, and for each alive CT compute an arrival time (nav path length $\div$ run speed) plus a defuse duration (5s with a kit, 10s without). The CT who would actually defuse is the one with the smallest finish time. Then

$$\text{defuse_time_margin} = \underbrace{t_{\text{fuse}}}_{\text{time left on bomb}} - \left(\underbrace{t_{\text{path}}}_{\text{nav path} \div \text{run speed}} + \underbrace{t_{\text{defuse}}}_{\text{5s with kit, 10s without}} \right), \quad (1)$$

which is positive if and only if a defuse is *physically attemptable*. The zero-crossing is a hard feasibility boundary in the game's physics. A refinement, `defuse_margin_kit`, computes the margin per CT with that CT's actual kit status and adds a T-interruption window (how long a T must survive to break the defuse).

Two features we built and rejected. We also computed (i) *map control in the neighbourhood of the live bomb* and (ii) a *dropped-bomb scramble* feature (who is closest to a loose bomb). Both were weak and redundant. We report them in Sect. 7.7 rather than quietly dropping them, because the *reasons* they fail are informative and generalise.

4.4 Pillar 4, Firepower, four ways (player skill prior)

A pre-round skill prior from HLTV per-player statistics, joined on (steamid, year), keyed by year because skill drifts season to season and a 2024 rating is not a 2026 rating. The statistics themselves are fixed; what varies is how they are *aggregated* from players to a side. We evaluate four encodings, each addressing one defect of the one before it, which is what makes the comparison clean.

- (a) **Summed rating, 2 features.** The obvious encoding, and the one the pillar started as: add each alive player’s HLTV Rating, once per side. Dense and simple. It is also, as Sect. 7.8 shows, almost entirely a re-encoding of *how many players are alive*: professional ratings live in a narrow band around 1.0, so a sum over n players is dominated by n .
- (b) **Mean rating, 2 features.** The direct repair: divide by the number of alive players, giving a genuine per-player average. The divisor is the number of alive players *that have an HLTV entry*, not the headcount. Players absent from the table never entered the sum, so dividing by the headcount would report a low “skill” for a side that merely has poor database coverage. This distinction is not cosmetic: on the 2026 holdout the implied per-player rating has median 0.84 under the headcount divisor and 1.09 under the correct one.
- (c) **Mean $\times$ team-rank weight, 2 features.** An individual rating does not say what calibre of opposition it was earned against: 1.05 on the top-ranked team is not 1.05 on the thirtieth. This encoding scales the mean by a team-ranking weight $w = 1/\log_2(\text{rank} + 1)$, so a player on the top team contributes at full weight ($w = 1$) and one on a rank-30 team at roughly a fifth ($w \approx 0.20$). Rank is resolved through a player-to-team-per-year table and an HLTV team-ranking table; players with no team record receive a rank-35-equivalent weight. We also test $1/\text{rank}$ and a linear $(31 - \text{rank})/30$.
- (d) **Mean + situational gates, 20 features.** Rather than one aggregate number per side, condition on what each player can actually be doing. This encoding is *side-aware* (separate CT- and T-side Rating, Firepower, Entrying, Trading and Opening statistics, since a player’s CT- and T-side output genuinely differ) and applies *per-player situational gates*: a lone survivor contributes his Clutching statistic and has Entry/Trading suppressed; a player with teammates alive contributes Entry and Trading; Opening statistics apply only when both sides are still 5-alive and an opening duel is actually possible. It adds an AWP-holder “sniping” role flag and a grenade-value-weighted utility term. All averaged, so the count confound of (a) does not return through the back door.

Sect. 7.8 reports the ablation across all four on a common skill-free base. The result previews the pillar’s fate: cross-validation rewards every encoding and the out-of-time test rejects every encoding. Throughout the rest of the paper the firepower-bearing sets (**F**, **EF**, **EFB2**) carry the *summed*, situationally-gated variant, the encoding the pillar was originally built with; Sect. 7.8 also re-tests the out-of-time conclusion with the cross-validation-preferred encoding, and it does not change.

Why this pillar is structurally different, and why that turned out to matter. Pillars 1–3 are computed *from the demo itself*: every input is present in the observation being scored. Pillar 4 is not. It requires an **external database, indexed by era**, and is therefore the only pillar that can be *unavailable at inference time*. In-sample this looks like a minor engineering detail. It is the single most consequential design decision in the paper (Sect. 7.15).

5 Models

Nine architectures, all evaluated under an identical protocol (Sect. 6).

5.1 Classical (5)

Logistic regression (standardised inputs, L2 penalty); random forest (300 trees); and the gradient-boosted decision-tree implementations XGBoost [2], LightGBM [6], and CatBoost [10].

All three boosters are tuned to *shallow, heavily regularised* trees (depth 3, $\lambda = 10$, subsample 0.8). **This matters more than any feature pillar.** Our initial depth-6 configuration overfit badly; fixing it was worth +0.007 AUC, larger than the contribution of *any* feature pillar in this paper. We report it as a warning: a poorly-regularised booster will happily swamp a genuine feature effect, and if we had run the map-control ablation at depth 6 we would have concluded that map control does nothing.

5.2 Deep (3)

All trained on GPU (NVIDIA B200, PARCC Betty cluster).

- **Causal TCN** (Temporal Convolutional Network [1]), a convolutional sequence model. The round is a sequence of per-second feature vectors, and dilated causal convolutions (dilations 1/2/4) ensure the prediction at second t attends only to $t' \leq t$: no within-round look-ahead. Sequence-to-sequence, one probability per second. This model can in principle learn *momentum*, which the per-snapshot models can only see through our hand-built trend and volatility features.
- **Causal Transformer encoder.** Same input, causal self-attention mask, sinusoidal positional encodings.
- **GAT (player graph)** [14]. A graph attention network that consumes the *raw per-player* state, one node per player carrying position, velocity, view angle, health, armour, equipment, kit, and side, for up to 10 alive players. The other models are not blind to health or equipment; the economy pillar already gives them *team-aggregated* totals (team health, total equipment value, players alive). The distinction is that the GAT sees these quantities *per player and with spatial position*, rather than as team sums, so it alone could in principle exploit which specific player, in which location, holds what. Masked multi-head self-attention over the player set, permutation-invariant, is masked mean-pooled to a round-state embedding. If any architecture were going to discover spatial structure our engineered features miss, it is this one.

5.3 Ensemble (1)

Soft-vote (mean predicted probability) over classical and deep members. A logistic *stack* was also tried and rejected (Sect. 7.10).

6 Evaluation Protocol

6.1 Cross-validation

5-fold **GroupKFold by match**. We never split a match across folds. Rounds within a match are strongly correlated (same players, same tactics, same map side, same day) and folding by *round* leaks that correlation across the train/test boundary and inflates AUC. All in-time numbers in this paper are out-of-fold (OOF) predictions.

6.2 Metrics

Primary (literature-comparable). AUC, log-loss, Brier score.

Complementary. ECE (10-bin), Brier Skill Score, and **contested-AUC**: AUC restricted to snapshots with equal players alive *and* $|\Delta\text{equipment}| \leq \1500 , 12.2 % of all snapshots.

Extended. The Murphy decomposition of the Brier score into reliability / resolution / uncertainty; sharpness; calibration slope and intercept (bin-free, from a logistic recalibration fit); an adaptive equal-mass ECE and a Kolmogorov–Smirnov calibration error (both binning-free, since 10-bin ECE is sensitive to bin placement); and a trajectory-level extreme-path diagnostic that benchmarks the forecast’s comebacks and write-offs against the calibrated-martingale law (Sect. 7.11).

6.3 Uncertainty and significance testing

We quantify uncertainty with a match-level block bootstrap ($B = 500$). Matches, rather than snapshots, are the independent resampling unit, because rounds within a match are strongly correlated; treating the 476,595 snapshots as independent would understate every interval by more than an order of magnitude. Each bootstrap iteration draws the set of matches with replacement and recomputes the metric on the fixed out-of-fold predictions, yielding percentile 95% confidence intervals.

Comparisons between two feature sets or two models use a *paired* bootstrap: on each resample we evaluate both configurations on the same set of matches and record their difference, and we report the 95% interval of that difference. A feature set is deemed to improve on the baseline when the paired interval of the AUC difference excludes zero. This paired construction is essential, because the two configurations produce strongly correlated errors, and the interval of the difference is therefore substantially narrower than a comparison of the two marginal intervals would suggest. Overlap of two marginal confidence intervals does not imply that their difference is non-significant; the paired difference is the correct test [3]. Under this test the pillar contributions relative to the economy baseline are significant on AUC, whereas the nine architectures are mutually indistinguishable (Sects. 7.3, 7.10). DeLong’s test corroborates the AUC differences, and the deep models additionally report a multi-seed mean and standard deviation. Significance is established formally on AUC; the proper scoring rules (log-loss, Brier) improve monotonically with it but are reported as point estimates with marginal intervals.

6.4 Out-of-time evaluation

The 2026 holdout is evaluated exactly once. No model, feature, or hyperparameter decision was made after observing it, because the evidential value of a held-out evaluation is contingent on that discipline. Any subsequent re-evaluation, such as after an updated skill database becomes available, is reported as a separate and explicitly disclosed result rather than combined with the original.

7 Results

7.1 The model matrix

Figure 3 gives the full model $\times$ feature-set grid. In sample, the best single model is **logistic regression on all pillars** (EFB2): AUC 0.8519 (95 % CI 0.8451–0.8582). The best overall is the four-model **soft-vote ensemble**: 0.8531 (0.8460–0.8598), which also has the best calibration (ECE 0.009, BSS 0.378). The economy baseline is 0.8465. Every pillar’s lift is small in pooled AUC but statistically significant, in the sense that its bootstrap CI excludes zero.

That *logistic regression* wins is itself informative: once the features are built, the aggregate signal is close to linear, and the non-linear models spend their capacity on variance rather than bias.

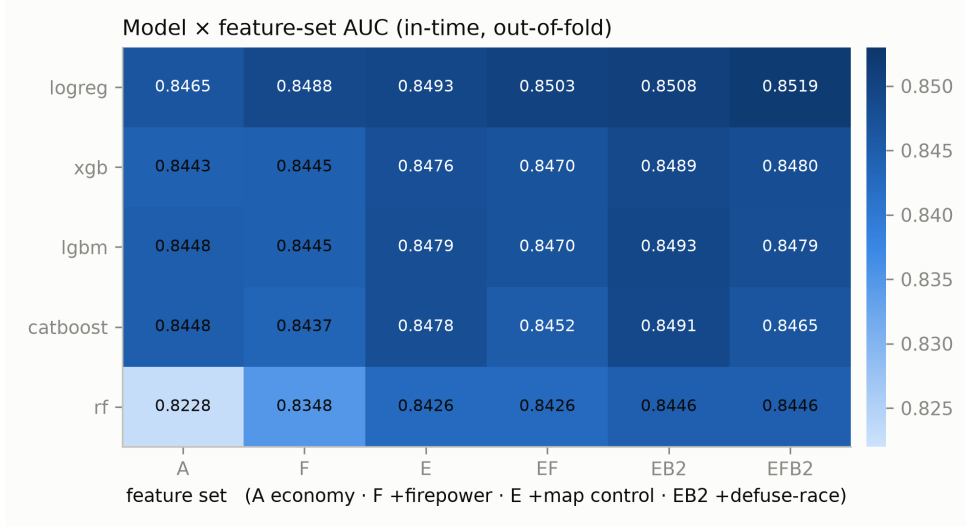


Figure 3: Model $\times$ feature-set AUC (in-time, out-of-fold). The grid is strikingly flat: the spread across nine architectures is smaller than the spread across feature sets, and both are small.

7.2 How much of the model is economy? The no-economy ablation

Every feature set in Fig. 3 contains the economy block, and the residual analysis of Sect. 7.4 measures what the other pillars add *after* economy. The complementary question is how far each block gets *alone*. Two definitional points shape the answer. The economy block is money (equipment value, economy class, armour, defuse kits; 7 columns) together with combat state (players alive, health, bomb planted, round time; 6 columns) and score (4 columns). And headcount leaks into the other pillars through columns that count alive players (per-zone player counts, AWP presence, counter-terrorists near the bomb; 14 columns), while the buy leaks through grenade counts and armour. We therefore split the economy block into its three parts, build variants of the non-economy sets with the count-type columns removed, and report every set on the equal-alive subset, where headcount carries no information.

Table 1 and Fig. 4 give the result for logistic regression; XGBoost is within 0.02 in every cell (Table 13). Money alone reaches an AUC of 0.830, 94 % of the production model’s above-chance discrimination, and it is the only block that is informative at the first second of the round: its AUC is 0.69 in the first five seconds, when every other block sits at chance. The non-economy pillars together reach 0.822 (92 %), but they do so late: they sit at 0.67–0.68 in the first ten seconds, which is the buy leaking through utility and armour, and approach the economy block only after 40 s. Combat state alone ends the round at 0.949, above money (0.933): in the last thirty seconds the round is read from who is alive, not from what was bought. Removing the money block from the production set costs only 0.010 AUC, and the resulting set transfers to 2026 without degradation (0.841 $\rightarrow$ 0.841), because grenade counts, armour and AWP presence re-encode most of the buy. Removing the fourteen count-type columns from the spatial block costs 0.017; on the equal-alive subset every non-economy set sits at 0.68 against 0.71 for the economy block, and combat state alone falls to 0.57, so the “combat” signal is headcount. Contested-AUC does not exceed 0.60 for any set. All paired match-level bootstrap intervals exclude zero, and every set transfers to the 2026 holdout within 0.02 of its in-time value with the ordering unchanged (one disclosed scoring).

The reading is that economy dominates the pooled numbers because it is the only information that exists before anyone moves, not because the spatial and bomb pillars are weak: given time, they recover nine tenths of what the full model knows, and the two families are largely redundant with each other.

Table 1: No-economy ablation (logistic regression, 5-fold out-of-fold). AUC with match-level bootstrap 95 % interval, the single disclosed 2026 scoring, AUC on the equal-alive subset, and contested-AUC. The number of columns is in parentheses.

Feature set	AUC	95 % CI	2026	equal alive	cont.-AUC
EB2, production set (72)	0.851	0.844–0.857	0.847	0.715	0.596
A, economy block (17)	0.847	0.840–0.853	0.842	0.710	0.587
Everything except money (65)	0.841	0.834–0.848	0.841	0.695	0.579
Money only (7)	0.830	0.822–0.838	0.823	0.693	0.536
All non-economy pillars (55)	0.822	0.815–0.828	0.822	0.679	0.567
Voronoi + tactical (43)	0.818	0.811–0.824	0.815	0.677	0.557
All non-economy, no count columns (41)	0.806	0.799–0.813	0.802	0.677	0.550
Combat state only (6)	0.797	0.790–0.804	0.787	0.571	0.578
Voronoi control only (9)	0.723	0.717–0.729	0.703	0.585	0.538
Bomb geometry + defuse race (18)	0.686	0.680–0.691	0.675	0.539	0.540
Round clock only (1)	0.555	0.549–0.560	0.539	0.535	0.548
Score and round number (4)	0.536	0.520–0.553	0.527	0.536	0.461

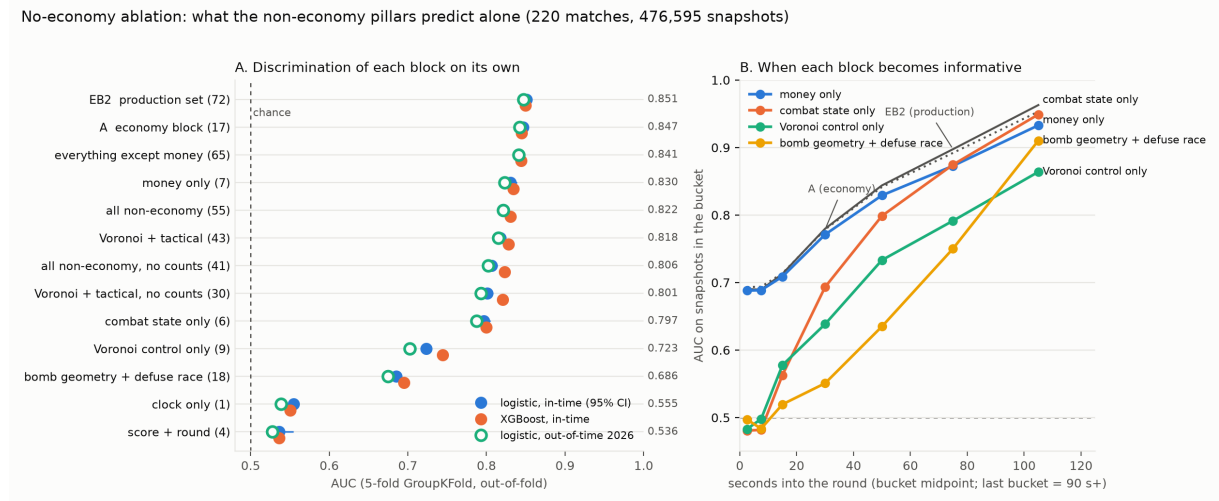


Figure 4: No-economy ablation. **Left:** out-of-fold AUC of each block on its own (logistic with 95 % match-level bootstrap interval; XGBoost; the single 2026 scoring). **Right:** AUC by seconds into the round for the four single blocks, with the economy block and the production set as references. Money is the only block informative at freeze-end; combat state, map control and bomb geometry start at chance and overtake money as the round unfolds.

7.3 Where the signal actually lives: contested-AUC

Pooled AUC hides the model’s real behaviour. Conditioning on how *even* the round is:

Table 2: AUC by round subset (XGBoost, out-of-fold).

Subset	Share of snapshots	A (economy)	EB2
All snapshots	100 %	0.844	0.849
Equal players alive	52 %	0.705	0.711
Even economy	,	0.683	0.689
Contested (both)	12.2 %	0.580	0.586

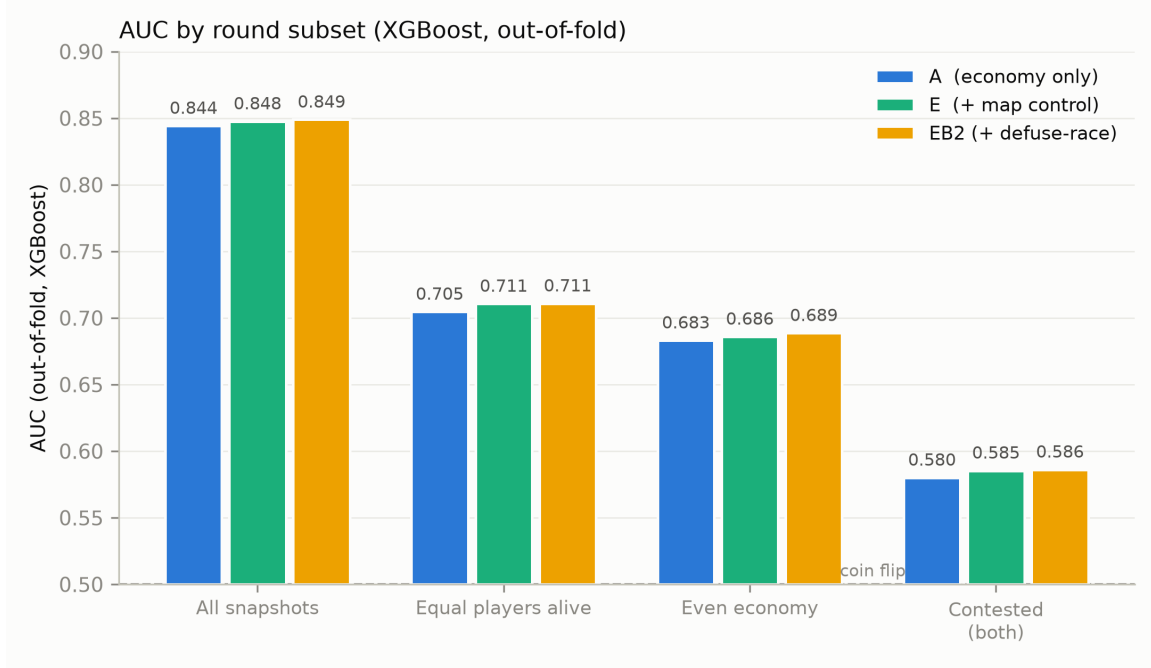


Figure 5: The headline AUC is carried by lopsided snapshots; in genuinely even rounds every model falls toward a coin flip.

The economy baseline collapses from 0.844 to 0.580, barely above a coin flip. The headline number is carried by lopsided snapshots (5-v-2, eco vs full-buy) that any model gets right, and which dominate per-second data precisely because a lopsided round *stays* lopsided for many consecutive seconds. In the states a viewer actually cares about (the round is even, the outcome is genuinely live) *no model in this study exceeds 0.60*.

We regard this as the honest characterisation of the state of the art. It does not mean the models are useless: correctly saying “this 5-v-2 is over” is what a broadcast overlay spends most of its time doing. It means the number conventionally reported is close to a measure of *how often the game is lopsided*, and reporting it alone invites the reader to over-estimate what the model knows.

The spatial pillars provide a consistent lift throughout, slightly larger in the contested regime. Their aggregate contribution is modest ($\approx +0.003$ – 0.005 AUC) but robust and significant across all classical architectures. We do not claim the spatial features are especially valuable in contested rounds, because the contested lift is not reliably larger than the pooled lift. The next two subsections examine the beyond-economy signal and the contested ceiling directly.

7.4 Beyond economy: a residual analysis

The nested ablation reports the AUC gained by adding a pillar, but it does not separate information that is genuinely new from information that re-encodes economy. We isolate the orthogonal component with a residual analysis. For each model we fit economy alone out-of-fold, form the economy logit $z = \text{logit}(p_A)$, and compare a model on z alone (which recovers economy) against a model on $[z, X_{\text{spatial}}]$, where X_{spatial} is the map-control, tactical, and bomb block (55 columns, no economy and no firepower). The added features can only act through the residual that economy leaves, so their contribution is the beyond-economy signal.

The contribution is significant and, importantly, transfers out-of-time. The paired match-level bootstrap interval of the AUC difference excludes zero for all four models both in cross-validation (range $+0.0035$ to $+0.0061$) and on the 2026 holdout (range $+0.0024$ to $+0.0048$) (Fig. 6, left;

Table 3: Beyond-economy contribution: AUC gain from adding the spatial/bomb block on top of the economy logit, with paired match-level bootstrap 95% intervals. All intervals exclude zero, in-time and out-of-time.

Model	in-time Δ AUC (95% CI)	out-of-time Δ AUC (95% CI)
Logistic	+0.0035 (+0.0018, +0.0053)	+0.0048 (+0.0009, +0.0087)
XGBoost	+0.0056 (+0.0038, +0.0076)	+0.0033 (+0.0011, +0.0058)
LightGBM	+0.0061 (+0.0045, +0.0078)	+0.0033 (+0.0010, +0.0053)
CatBoost	+0.0051 (+0.0035, +0.0067)	+0.0024 (+0.0001, +0.0046)

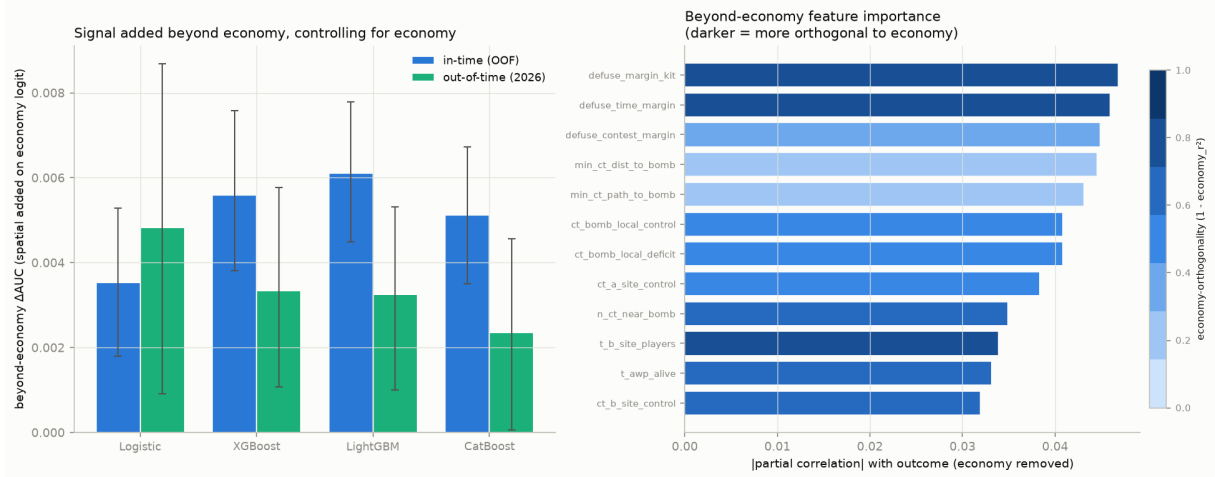


Figure 6: Residual analysis. **Left:** AUC gained by adding the spatial/bomb block on top of the economy logit, per model, in-time and out-of-time, with paired bootstrap intervals; all exclude zero. **Right:** beyond-economy feature importance (partial correlation with the outcome after removing economy), shaded by orthogonality to economy. The defuse-race margins are both the strongest and the most orthogonal.

Table 3). The contested-AUC difference, in contrast, is not significant (for logistic regression, +0.005 with a 95% interval of -0.015 to $+0.025$): the beyond-economy signal helps in partially decided rounds, not in even ones, consistent with the ceiling analysis in Sect. 7.5.

A Frisch-Waugh-Lovell orthogonalisation identifies which features carry that signal. Partialling economy out of both the outcome and each feature and ranking by the partial correlation of the residuals places the two defuse-race margins at the top, and these have the lowest economy R^2 (approximately 0.15), meaning they are largely orthogonal to economy and carry genuinely new information (Fig. 6, right). Features such as the nearest counter-terrorist’s distance to the bomb rank high in raw importance but have a high economy R^2 (approximately 0.69), indicating that much of their apparent value is economy re-encoded.¹ This mirrors the firepower count confound of Sect. 7.8 and confirms the defuse-race feature as a source of orthogonal, transferable signal.

7.5 The contested-round ceiling

The contested-AUC of approximately 0.58 raises a question the pooled number cannot answer: is it low because the features are weak, or because even-round outcomes are close to irreducibly

¹This is not because money predicts distance. Recall (Sect. 4.1) that the “economy” pillar includes combat-state variables such as players alive, team health, and round time. Those strongly constrain the nearest-CT distance: with five counter-terrorists spread across the map the closest one is usually near the bomb, whereas with one alive it can be anywhere, and once the bomb is planted the distance is measured to a fixed site. Fig. 2 shows this directly, its points coloured by the number of counter-terrorists alive. The distance is therefore largely a function of headcount and round phase, which the economy pillar already carries.

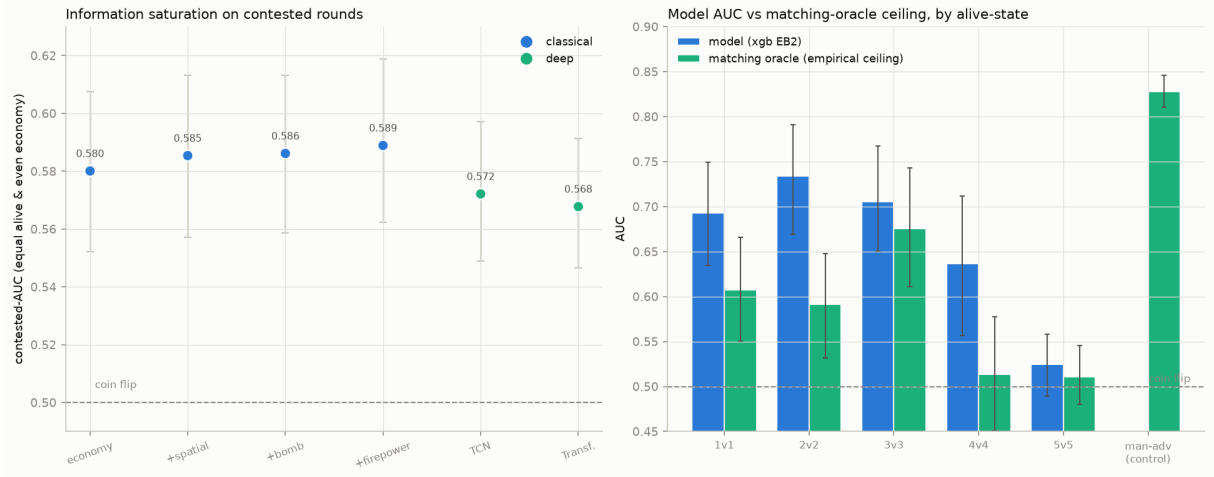


Figure 7: The contested-round ceiling. **Left:** contested-AUC is flat near 0.585 across representations of increasing richness, including deep sequence models. **Right:** model AUC and a model-free matching-oracle ceiling by alive-state; both fall to a coin flip at 5-versus-5 even (71% of contested snapshots), while the man-advantage control reaches 0.83, validating the estimator.

random? We address it three ways, all at the snapshot level, since the round-level contested slice is underpowered.

First, contested-AUC does not improve with representation richness. Across economy, +spatial, +bomb, and +firepower it remains near 0.585, and the causal sequence models are slightly lower (Fig. 7, left). Adding information does not break the ceiling.

Second, the ceiling is concentrated in one state. Decomposing the contested set by headcount, the 5-versus-5 even state has an AUC of 0.525 with a counter-terrorist win rate of 0.483, close to a coin flip, and it constitutes 71% of contested snapshots (41,338 of 58,368). Smaller even states, where positional and bomb information has accumulated, are more predictable (1-versus-1 to 3-versus-3 reach 0.69 to 0.73). The pooled contested-AUC is therefore dominated by the symmetric, pre-engagement phase of the round.

Third, a model-free matching estimate confirms that this is a property of the game rather than of the model. For each even snapshot we locate its nearest neighbours in a 60-dimensional observable-state space, drawn from different matches so that they do not share the round label, and measure how often near-identical states end differently. For 5-versus-5 even states the disagreement rate is 0.495, indistinguishable from a fair coin, and a predictor with perfect knowledge of the local outcome frequency reaches an AUC of only 0.511 (Fig. 7, right). On a one-player man-advantage control the same estimator returns a disagreement rate of 0.314 and an oracle AUC of 0.828, confirming that it detects strong signal where the state genuinely determines the outcome. The near-coin-flip verdict for the symmetric opening is thus a characterisation of the game: at the start of an even round, the observable state does not determine who wins.

7.6 Map control: realistic $\neq$ predictive (unless stabilised)

This is the central spatial result. Recall Fig. 1: the grey model is visibly the truer description of the game. Here is what it buys us.

Table 4: The map-control ablation (XGBoost, out-of-fold). The most physically faithful representation is the least predictive.

Representation	AUC	Verdict
Economy only (A)	0.8443	baseline
+ Voronoi (proximity)	0.8468	best single control model
+ Grey (LOS + FOV + smoke)	0.8451	<i>most faithful, least predictive</i>
+ Territory (grey + 15 s memory)	0.8453	memory restores it

The instantaneous LOS/FOV/smoke model is a far better description of what the players can actually *see and contest*, and it is a **worse** predictor than naive proximity.

The reason is temporal. Yaw is twitchy: a player flicks his crosshair across 90 degrees in a fraction of a second, and the set of areas he “controls” under an instantaneous FOV test churns violently from tick to tick. Instantaneous control therefore *flickers*, while round outcomes depend on *stable* territorial state, who has *taken* banana, not who happens to be looking down it at this exact instant. Granting the same model a 15-second memory (cleared space stays yours until contested) recovers parity with Voronoi.

Predictiveness comes from temporal stability, not physical fidelity.

This carries a concrete methodological warning. A practitioner who builds a vision-based control surface, finds it does not beat a proximity baseline, and concludes that “map control does not predict outcomes” has drawn the wrong conclusion from the right experiment. The representation was too jittery; the concept was fine.

Voronoi and territory are largely redundant with one another ($r = 0.47$); stacking territory on top of Voronoi yields ≈ 0 additional AUC. We therefore retain Voronoi as the primary control feature, the crude model, kept for the un-crude reason that it is stable.

What the lift looks like in a single round. Aggregate AUC deltas of +0.003 are easy to dismiss, so Fig. 8 shows a round where the map-control pillar is doing something a human analyst would recognise. The economy-only model (grey) finishes the round convinced the CTs are winning, it spikes to 0.82, because on paper the CTs have the equipment. The map-control model (red) sees that the T side has actually taken the map and holds its prediction near 0.28, a −54 percentage-point correction. The T side won.

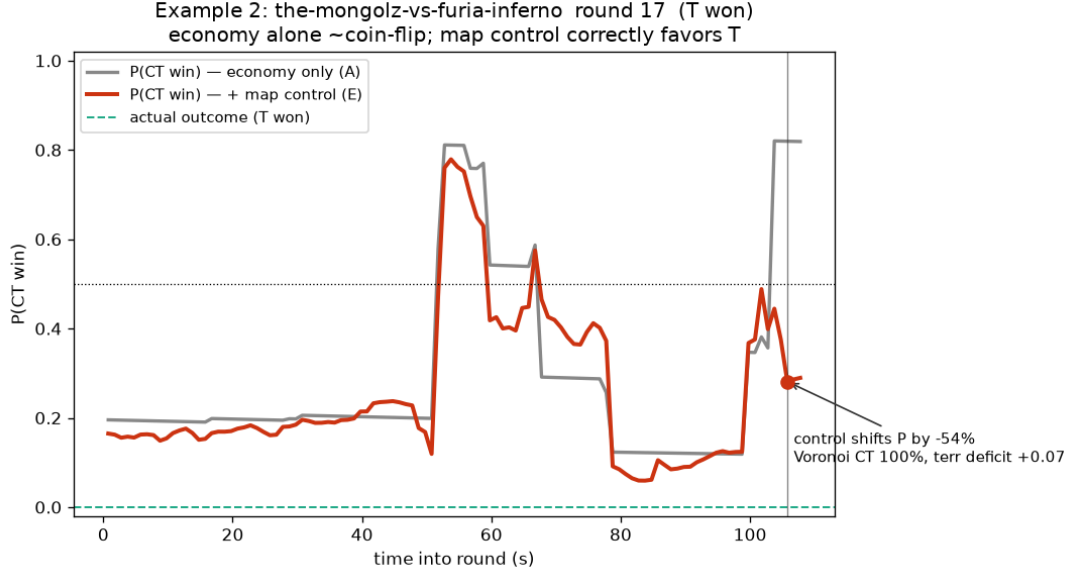


Figure 8: A round where map control earns its keep. Economy alone (grey) ends up near 0.82 for the CTs; adding map control (red) corrects this to ≈ 0.28 . The T side won. These corrections are individually large but rare, which is exactly why the pooled AUC lift is small.

That last point deserves emphasis, because it is a general property of this problem: *large, correct, individually convincing corrections that are nonetheless rare produce small pooled AUC gains*. This is another reason the pooled number is a poor lens.

7.7 Bomb geometry: the defuse race

Recall the defuse-race feature `defuse_time_margin` from Eq. 1: the fuse time remaining minus the time the nearest counter-terrorist needs to reach the bomb and defuse it. It is positive when a defuse is still physically possible and negative when it is not, so it encodes, in one number, whether the counter-terrorists can still win the post-plant. We measure each feature’s importance by *permutation importance*: we randomly shuffle that feature’s column, breaking its link to the outcome, and record how much out-of-fold AUC drops; a large drop means the model relied on it. By this measure `defuse_time_margin` is **rank #8 of 68**. The seven features above it are all part of the economy and combat-state baseline (both sides’ equipment value, both sides’ team health, players alive, armour), so the defuse margin is the strongest single feature *added on top of* that baseline. The residual analysis of Sect. 7.4 independently confirms it is also the most economy-orthogonal feature, i.e. the one carrying the most genuinely new information. Adding the bomb-live pillar (set EB2) is significant on *every* architecture (all bootstrap CIs exclude zero), and the gain is concentrated exactly where the physics says it should be:

- **post-plant log-loss falls 7–8 %** (XGBoost: $0.187 \rightarrow 0.172$; post-plant AUC $0.962 \rightarrow 0.967$);
- **endgame log-loss falls across all models** ($0.401 \rightarrow 0.396$);
- **calibration is preserved** ($ECE \approx 0.013$): the feature adds discrimination without distorting the probabilities.

Figure 9 shows the mechanism, and it is the cleanest single piece of evidence in the paper. The SHAP contribution is a **step function with its jump at exactly zero**, the feasibility boundary. When the margin is positive (a defuse is physically attemptable) the feature contributes ≈ 0 ; the instant it goes negative, the contribution drops to ≈ -0.8 and stays there. The model

has learned the game’s physics as a hard constraint: *when a defuse becomes impossible, abandon the CTs*.

Contrast this with the smooth, graded SHAP curve for raw distance-to-bomb (Fig. 2). Same underlying geometry; one encoding makes the boundary learnable and the other does not. **This is the whole argument for physically-derived features in one pair of plots**, and it is behaviour none of the deep models discovered on their own from the raw distances.

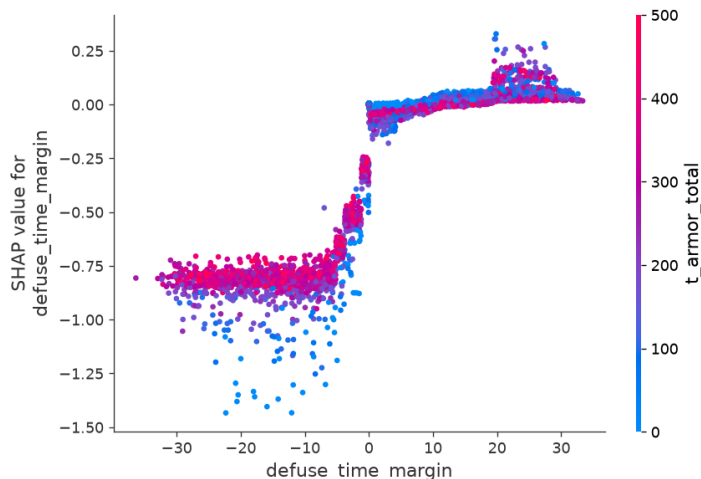


Figure 9: SHAP dependence for `defuse_time_margin`. The step at zero is the *feasibility boundary*: to the right a defuse is physically attemptable and the feature is near-neutral; to the left it is impossible and the model confidently writes the CTs off. The model has learned a hard physical constraint, not a soft correlation.

Negative results, reported. Neither *map control around the live bomb* nor the *dropped-bomb scramble* feature helped; both were weak and redundant. The reasons generalise:

- A loose (dropped) bomb is a *consequence* of a losing round, not a cause of one. The T carrying it died, and “a T died” is already in the model. The feature is downstream of information we already have.
- Bomb-neighbourhood ownership duplicates the existing site-control features, because the bomb is nearly always at a site.

The lesson is that what matters post-plant is the *race geometry* of Eq. 1, not local ownership of the ground the bomb happens to sit on.

7.8 Firepower: a confound, its repair, and what survives

Firepower is the weakest of the four pillars in-sample, and the most instructive.

The summed encoding, and the most seductive artifact in the study. Encoding (a)’s feature `ct_rating_sum - t_rating_sum` came out at **permutation-importance rank #1** of all 68 features. Taken at face value this is a triumph for the skill pillar: player quality is the single most important thing in Counter-Strike, exactly as folklore says.

It is nothing of the sort, and Fig. 10 shows why in one glance. The feature’s values **cluster at the integers 0, 1, 2, 3, 4, 5**. It is not measuring skill; it is *counting players*. Because rating is *summed over alive players* and professional ratings live in a narrow band around 1.0, the sum is dominated by how many terms are in it. The correlation with the plain player-count advantage is **0.987**, and the two predict the outcome identically ($r = 0.498$ each).

Table 5: Firepower encodings on a common EB2 (skill-free) base, contested-AUC. Left block: five-fold out-of-fold cross-validation. Right block: the 2026 out-of-time holdout. Every encoding improves on the skill-free model somewhere in cross-validation; none improves on it out-of-time. The ordering between the two blocks is close to reversed.

encoding	cross-validation					out-of-time (2026)				
	log	xgb	lgbm	cat	rf	log	xgb	lgbm	cat	rf
EB2 (no firepower)	0.596	0.586	0.590	0.588	0.586	0.655	0.628	0.637	0.650	0.577
(a) summed rating	0.598	0.593	0.593	0.591	0.587	0.631	0.628	0.612	0.635	0.564
(b) mean rating	0.596	0.588	0.588	0.582	0.589	0.644	0.631	0.628	0.644	0.571
(c) mean $\times$ team rank	0.594	0.604	0.603	0.599	0.599	0.656	0.637	0.641	0.645	0.564
(d) mean + gates	0.605	0.589	0.590	0.584	0.580	0.592	0.596	0.588	0.585	0.542

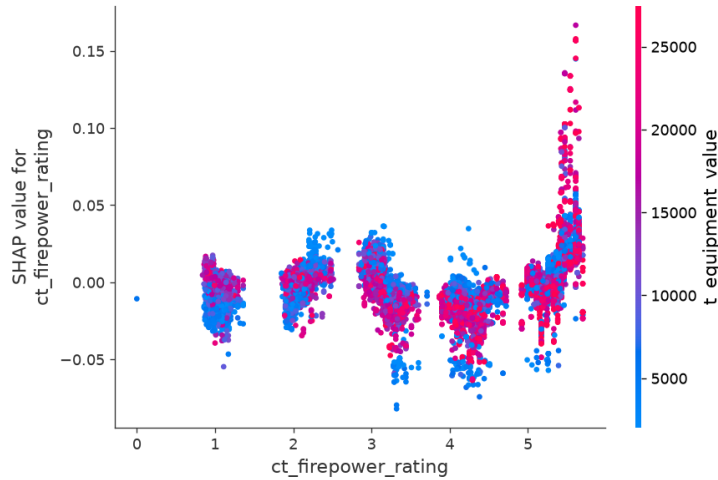


Figure 10: The count confound, made visible. The “team firepower rating” is a *sum* of per-player ratings over alive players, and professional ratings cluster near 1.0, so the feature’s values pile up at the integers 0–5. The top-ranked “skill” feature is mostly a re-encoding of *how many players are alive*, which the economy pillar already supplies. Its *marginal* contribution is ≈ 0 .

This is a cautionary tale about permutation importance on collinear features: it will rank a proxy above the thing it proxies for, and it will do so confidently.

The repair, and two directions from it. Encoding (b) is the direct fix: divide by the number of alive players with a database entry. It works, in the narrow sense that it does what it was built to do. Averaged over the five classical models, moving from (a) to (b) recovers two thirds of the pillar’s out-of-time damage, contested-AUC $-0.015 \rightarrow -0.006$ against the skill-free base. The confound is real and removing it is worth something. It is simply not worth enough: (b) is still, on average, *behind* using no skill prior at all.

From the repaired base we explored the two directions the pillar plausibly had left. Encoding (c) asks whether the rating needs adjusting for the calibre of the team it was earned on; encoding (d) asks whether one aggregate per side is too coarse and the statistics should be conditioned on what each player can currently be doing. Table 5 gives all four on a common skill-free base.

Cross-validation rewards every rung; the out-of-time test rejects every rung. Read the two blocks of Table 5 against each other. In cross-validation the mean gain over the skill-free base is $+0.003$ for (a), -0.001 for (b), $+0.011$ for (c) and $+0.001$ for (d); (c) is the best encoding on four of the five models and (d) on the fifth. Out-of-time the same four are -0.015 , -0.006 ,

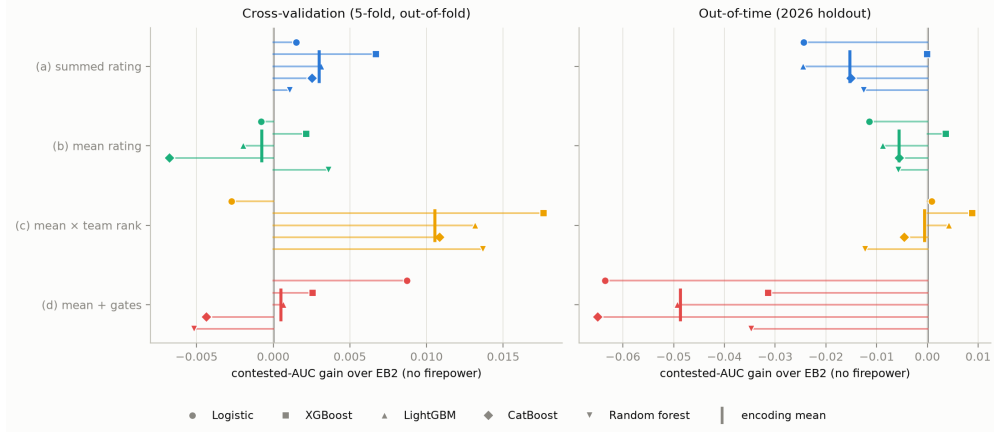


Figure 11: Contested-AUC gain over EB2 (no firepower) for each firepower encoding and model; the heavy tick is the mean over the five models. **Left:** five-fold cross-validation, where every encoding sits at or above the skill-free base and the situationally-gated (d) is the best on the linear model. **Right:** the 2026 out-of-time holdout, where the ordering collapses: (d) falls furthest on all five models and no encoding rises clearly above the base. **Note the axes:** the panels are drawn on their own scales because the out-of-time losses are roughly four times the size of any in-fold gain (-0.065 to $+0.010$ against -0.007 to $+0.018$). The reversal between the panels, not the level in either, is the finding.

-0.001 and -0.049 , and the number of models on which each beats the skill-free base is 0, 1, 3 and 0 out of five. The encoding that cross-validation likes most, (d), is the one that fails hardest out of sample, by a margin (contested-AUC -0.049 , on every one of the five models) far outside the bootstrap noise of ± 0.02 – 0.03 . Twenty sparse, frequently-`NaN`, situationally-gated features are exactly what a flexible model overfits, and the cross-validated metric cannot see it because every fold is drawn from the same era.

The conclusion does not depend on which encoding we chose. The obvious objection to a negative result about a pillar is that it was encoded badly. Selecting an encoding on out-of-time performance would be selecting on the test set, so we select on cross-validation, which prefers (c) by a clear margin ($+0.011$ contested-AUC, three times the next best). Re-running the out-of-time comparison with (c) leaves the conclusion intact: averaged over the five models it lands 0.0006 contested-AUC *below* the skill-free model, inside the noise in the unhelpful direction. The firepower-bearing sets elsewhere in this paper carry the summed, gated encoding, which Table 5 shows is not the strongest one; the finding of Sect. 7.16 is not an artifact of that choice, because the strongest one does not survive either.

Summary. Firepower carries a marginal signal over bare economy (the F versus A comparison is significant for logistic regression, $+0.0022$, 95% CI 0.0003 to 0.0045), but that signal vanishes once the demo-derived pillars are present. We can now say why, rather than only that. The pillar had one identifiable defect, a summed statistic that mostly counted players; removing it recovers two thirds of the out-of-time damage and still leaves the prior behind the skill-free model. Neither direction available from the repaired encoding rescues it: adjusting for opponent calibre lands on the baseline, and conditioning on situation is the worst option out-of-time despite being the best in cross-validation. **The pillar’s binding constraint is therefore not its encoding but its content:** the observable match state already absorbs whatever team-quality signal an external skill prior can provide. This is a stronger claim than a null result, because the encoding hypothesis was the obvious alternative explanation and it has been tested and ruled out.

7.9 What the models actually use

Figure 12 shows collinearity-robust permutation importance for set E across all five classical models. Two things are worth noting.

First, the top of the table is **economy**: equipment value and health for both sides, across every model. This is the honest picture, and it is consistent with the contested-AUC collapse (Sect. 7.3), since equipment and health are precisely the variables that go to zero in a lopsided round.

Second, look at `min_ct_dist_to_bomb`: it is the single largest feature for *logistic regression* (blue) and comparatively minor for the tree models. This is a useful reminder that “feature importance” is a property of the *model*, not of the world. The linear model leans hard on a monotone spatial feature because it cannot represent the interaction the trees construct instead.

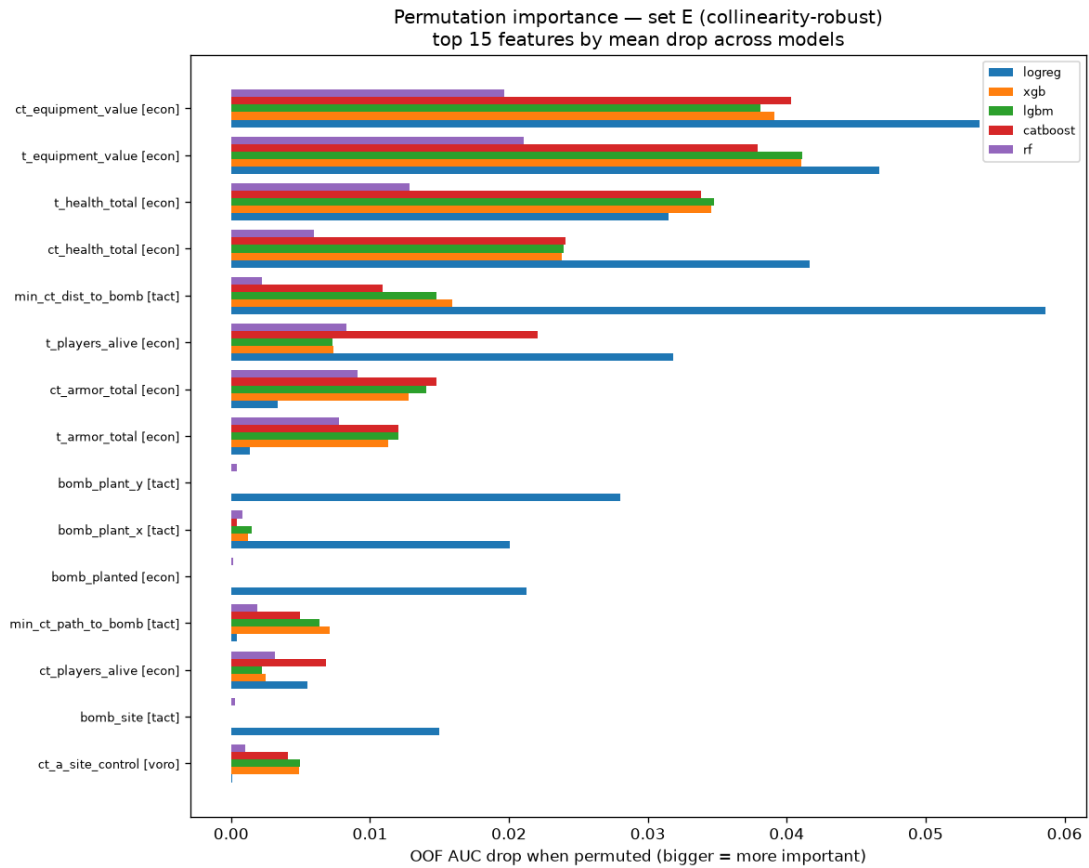


Figure 12: Permutation importance (OOF AUC drop when a feature is shuffled), set E, all five classical models. Economy dominates. Note that `min_ct_dist_to_bomb` is the top feature for logistic regression but minor for the trees: importance is a property of the model, not of the game.

7.10 Deep models are statistically indistinguishable from the classical baseline

Table 6: Neither the sequence models nor the graph model beats a calibrated logistic regression. All intervals overlap all point estimates.

Model	AUC	95 % CI
Ensemble (soft-vote)	0.8531	(0.8460, 0.8598)
Logistic regression (EFB2)	0.8519	(0.8451, 0.8582)
LightGBM (EB2)	0.8493	(0.8424, 0.8558)
TCN (causal, sequence)	0.8488	(0.8420, 0.8557)
Transformer (causal)	0.8473	(0.8398, 0.8541)
GAT (raw player trajectories)	0.8465	(0.8396, 0.8534)

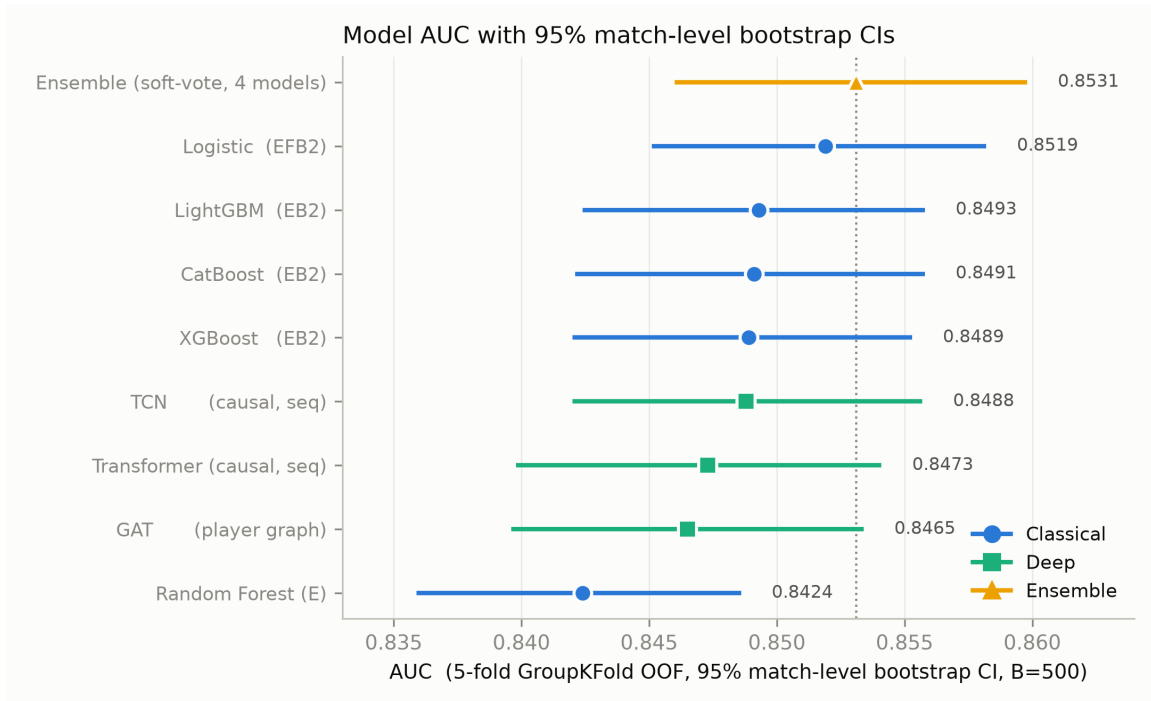


Figure 13: All nine architectures with 95 % match-level bootstrap CIs. Every point estimate lies inside the others’ intervals.

Every point estimate lies inside the others’ 95 % confidence intervals. Neither the sequence models (which can see *momentum*) nor the graph model (which sees the raw positions, velocities and view angles our aggregates throw away) beats a calibrated logistic regression.

Three observations sharpen this.

1. **Sequence length is critical** for the TCN: AUC 0.849 at 160 timesteps, 0.826 at 100, 0.780 at 64. Truncation discards the decisive late-round snapshots. A reader seeing only the 64-step number would wrongly conclude sequence models are hopeless here, and an author in a hurry might have published exactly that.
2. **Multi-seed variance is tiny** (0.8485 ± 0.0004). The tie is not an artifact of an unlucky initialisation.
3. **The deep models still earn their place in the ensemble.** The soft-vote of classical

+ deep beats either family alone, the textbook diversity benefit. A logistic *stack* matched the soft-vote’s AUC but destroyed calibration (ECE 0.042) and was rejected: for a live win-probability product a well-discriminating but badly-calibrated model is worse than useless, because the number on screen *is* the product.

Interpretation. The engineered features already encode what these architectures would otherwise have to learn (trend and volatility for the TCN, zone control for the GAT) and 220 matches ($\approx 4,900$ rounds) is a small corpus for a neural network. This is a statement about *this data scale*, not a claim that deep learning cannot help. We would expect the GAT in particular to pull ahead given an order of magnitude more matches, precisely because it is the only model with access to information the feature engineering discards.

The equivalence persists out-of-time. We also evaluated the TCN and Transformer on the 2026 holdout (they consume every feature column, so they are the deep analogue of the all-pillar EFB2 configuration). On the same-year holdout the TCN reaches AUC 0.8443 (95 % CI 0.825–0.862) and the Transformer 0.8448 (0.826–0.863), landing exactly on the classical EFB2 out-of-time number (0.8450, logistic) with fully overlapping intervals. The in-time tie is therefore not an in-sample artifact: it holds under distribution shift. Two smaller observations. First, the deep models degrade *less* out-of-time than the classical EFB2 (TCN -0.0046 , Transformer -0.0025 from their in-time OOF, vs -0.007 to -0.009 for the classical EFB2 models), they do not overfit the training era more, if anything slightly less. Second, because they ingest firepower along with everything else, they sit at the EFB2 level and thus *below* the no-firepower EB2 (0.847–0.850); the deep models inherit the skill-prior penalty of Sect. 7.16 and cannot shed it without retraining on a reduced feature set. The Transformer’s contested-AUC (0.62) is the highest of any model out-of-time, but the interval is wide (0.55–0.68) and we do not read it as significant.

7.11 Honest probabilities: calibration, phases, and comebacks

A win-probability model’s output *is* the product. Discrimination is necessary; calibration is what makes the number mean anything.

Figure 14 is the reliability diagram with bootstrap CIs. Both the linear and the tree model sit on the diagonal within error nearly everywhere (logreg EFB2 ECE 0.016, CI 0.013–0.023; xgb EFB2 ECE 0.013, CI 0.010–0.020). The Murphy decomposition shows the gain from adding pillars is *resolution* (discrimination), not reliability: the features add skill *without* costing calibration, an outcome one wants and is not guaranteed.

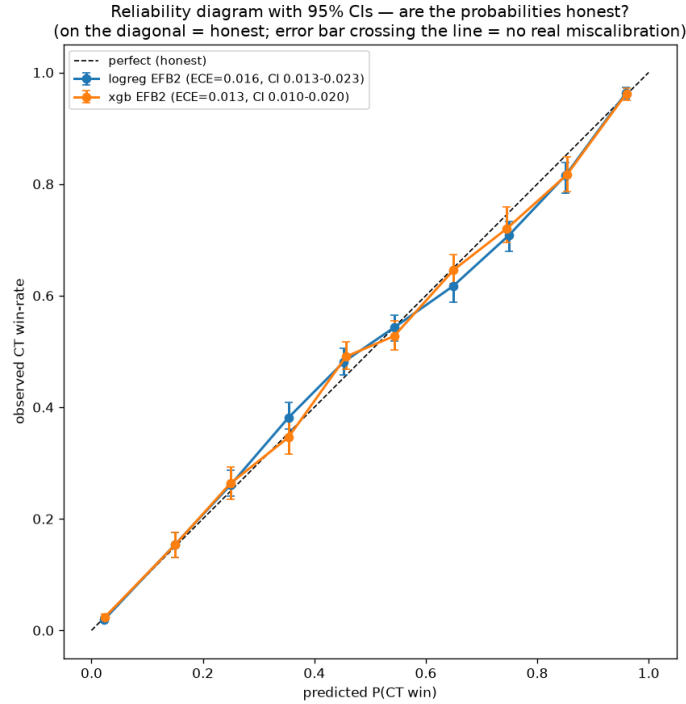


Figure 14: Reliability with 95 % bootstrap CIs. Error bars crossing the diagonal indicate no real miscalibration. Both models are honest.

Aggregate ECE can hide a model that is honest on average and dishonest at a particular phase of the round, so Fig. 15 breaks calibration out by time into the round. Both models stay near or below the 0.02 “well-calibrated” line throughout, and both are *best* at the end (30s+), reassuring, since that is when the model is most confident and most consequential. The linear model is slightly worse mid-round (25–30 s), which is where tactical situations are most complex and least linear.

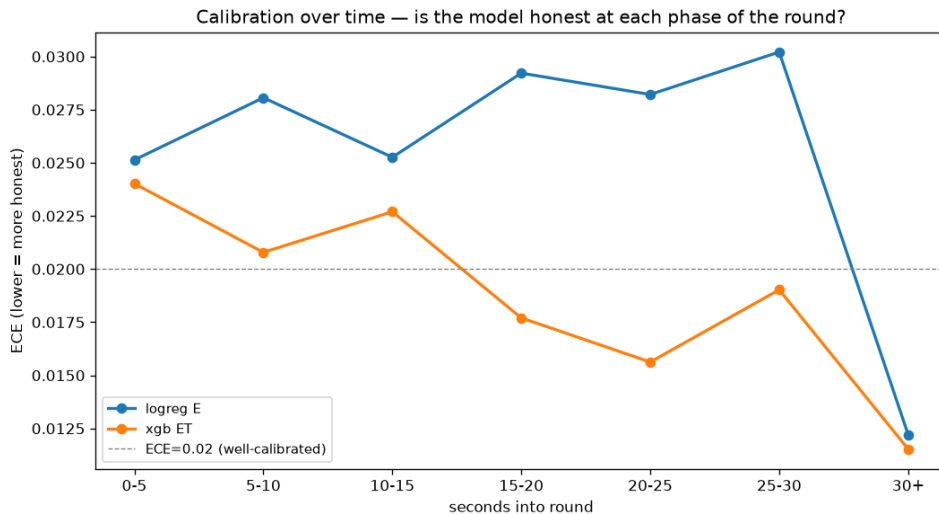


Figure 15: Calibration by phase of round. Both models are honest at every phase, and most honest at the end.

Trajectory honesty: are comebacks as surprising as they look? A pointwise-calibrated model can still misrepresent the *drama* of a round. How often the forecast swings to near-

certainty, or writes a side off, is a property of the whole path, not of any single second, and a write-off is the most consequential thing a live overlay says. Under ideal sequential calibration the per-second forecast p_t is a bounded Doob martingale that begins at the round’s freeze-end probability p_0 and ends at the outcome, and the extremes of such a path obey an exact law [9, 8]: the probability that the eventual loser ever reaches x is $\frac{p_0}{1-p_0} \cdot \frac{1-x}{x}$, which fixes how often the eventual winner should be written off below a threshold.

We measure both the pointwise tails and the path extremes. On the tails, when the model says 5 % the side wins 0.7 % of the time, and when it says 10 % it wins 6.8 %. The eventual winner is written off (≤ 10 %) at some moment in 7.4 % **of rounds**, against a continuous-martingale benchmark of 11.1 %; the same ordering holds at every threshold (observed 2.9 %, 7.4 %, 18.8 % at $\leq 5, 10, 20$ %, versus benchmarks of 5.3 %, 11.1 %, 23.9 %; Fig. 16, left). A one-sided Kolmogorov–Smirnov test on the probability-integral transform of the loser’s-peak statistic finds *no* upper-tail inflation ($D^{\text{upper}} = 0$; Fig. 16, right).

Two conclusions follow. First, the model does not over-react: it never assigns extreme confidence to eventual losers more often than an honest forecast would, in contrast to published feeds that do (the same diagnostic flags ESPN’s NBA win-probability series as over-confident [9]). For a live overlay this is the consequential direction, and the model does not cry wolf. Second, the write-off rate sits *below* the continuous benchmark, which is precisely the conservatism a discrete, jump-driven forecast must show: a round’s probability moves only at kills, so it has far fewer opportunities to reach an extreme than a continuously diffusing path, and the discrete benchmark is a conservative lower reference for exactly this reason [9]. A viewer who sees the curve dip to 10 % and then witnesses a comeback about one round in fourteen is seeing an honest, non-sensational number.

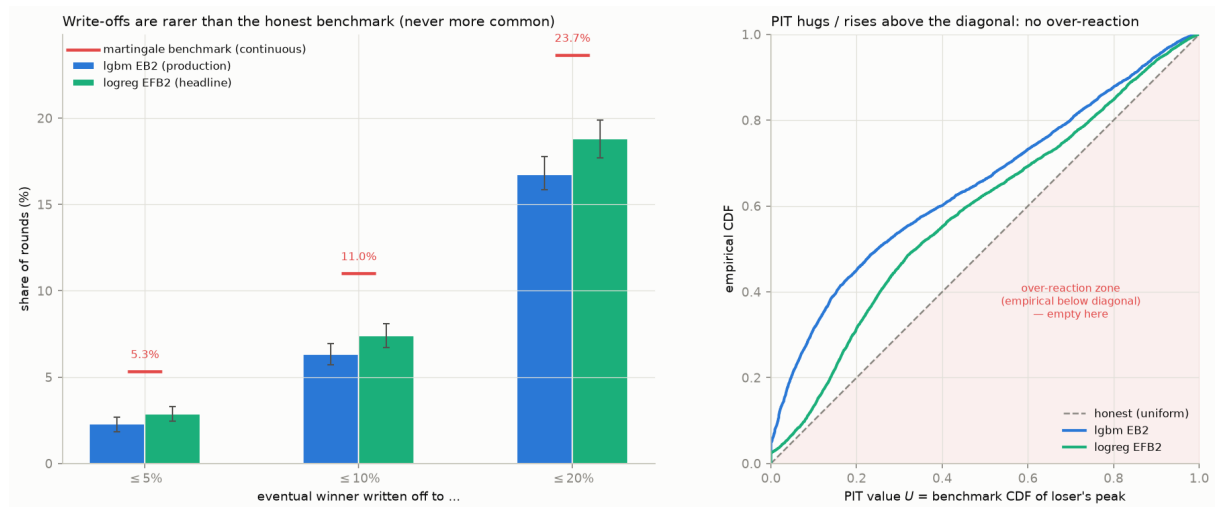


Figure 16: Trajectory-level calibration. **Left:** how often the eventual winner is written off to $\leq 5/10/20$ %, observed (bars, 95% match-level bootstrap CIs) versus the calibrated-martingale benchmark (red). Observed sits below the benchmark at every threshold, and never above it. **Right:** the probability-integral transform of the loser’s-peak statistic. Over-reaction would push the empirical curve into the shaded region below the diagonal; the CS2 curves sit on or above it, so there is no upper-tail inflation, only the mild conservatism expected of a discrete forecast.

7.12 The round-start prior

The trajectory benchmark above anchors each round at p_0 , the forecast at freeze-end, which so far has been the per-second model’s own first value. Two questions follow: what *should* the forecast be before anyone moves, and is the per-second path consistent with its own starting point? Both are answered at the round level (4,866 rounds; 5-fold GroupKFold by match; paired

match-level bootstrap, $B = 500$). The 2026 holdout is not used in this subsection; the results are in-time.

What is known at freeze-end. Table 7 is a nested ladder of round-level logistic models. Format and score alone give an AUC of 0.536; the buy state (equipment, armour, kits, utility, AWP) raises it to 0.682 and is the whole of the round-start signal. Recent history (previous result, loss streak, previous equipment) adds +0.004. External priors add +0.010 in the same-year construction, which is leaky because season ratings are computed partly from the matches predicted, and +0.005 with a confidence interval that includes zero in the honest lagged construction. Priors alone are at chance. A dedicated round-level refit of the economy columns (0.691) is no better than the per-second model’s own first value (0.692), so the appropriate construction is a stack: the first-snapshot logit of the per-second economy model as an offset, plus history and lagged priors (0.695; 0.700 with the leaky priors). This is the same conclusion as the firepower pillar, reached at the one point in a round where a prior has no in-round state to compete with: in this game, pre-round information beyond the buy is nearly uninformative about a single round.

Table 7: Round-start prior: nested logistic ladder at the round level (5-fold out-of-fold). The last column is the paired match-level bootstrap difference against the buy-state rung. “Lagged” priors use the previous season and are undefined for the 2024 matches, for which no 2023 tables exist.

Rung	columns	AUC	Δ AUC vs buy state (95 % CI)
Format, side, score	9	0.536	−0.146 (−0.165, −0.128)
+ buy state	37	0.682	—
+ recent history	45	0.687	+0.004 (+0.000, +0.009)
+ team and player priors, same year (leaky)	54	0.692	+0.010 (+0.004, +0.016)
+ team and player priors, lagged	58	0.688	+0.005 (−0.000, +0.012)
Priors only, lagged	15	0.504	−0.179 (−0.201, −0.154)
Per-second economy model, first snapshot	17	0.692	+0.010 (+0.003, +0.016)
Stack: first-snapshot logit + history	9	0.695	+0.013 (+0.006, +0.019)
Stack + lagged priors	22	0.695	+0.013 (+0.005, +0.021)
Stack + same-year priors (leaky)	18	0.700	+0.017 (+0.010, +0.024)

Is the path consistent with its start? Anchoring the per-second EB2 curve at the demo-derived prior, two exact identities are checked. For a calibrated prior, $E[(Y - p_0)^2] = E[p_0(1 - p_0)]$; the observed ratio is 1.02, and the per-second model’s first value differs from p_0 by +0.001 on average, so the prior is calibrated and the curve starts where it should. For a martingale, $(Y - p_0)^2$ equals the initial jump squared plus the summed squared increments plus the final jump squared, in expectation; the observed ratio is 1.31 for EB2 and 1.31 for the economy-only model. The per-second path therefore carries about 30 % more squared movement than an honest sequential forecast that starts at p_0 and ends at the outcome. The excess is short-range mean reversion: increment autocorrelations are −0.04, −0.03, −0.03, −0.01 at lags of one to five seconds and vanish beyond ten. It is not noise to be smoothed away: shrinking every increment toward p_0 by a factor 0.85 restores the identity exactly and costs 0.011 in log-loss, so the movement is informative. The model is calibrated at each instant but not consistent with its own history; the remedy is a path-aware forecast (lagged values of p_t as inputs, or the sequence models of Sect. 7.10 scored on the same identity), which we leave to future work. The pointwise extreme-path test above did not see this because it examines the maximum of the path, not its total variation.

The first kill, and the stronger team. The first kill of a round (median at 33s) moves the curve by about 0.2 in even states. The reaction to a counter-terrorist death is calibrated

(post-kill mean 0.216 against a realised CT win rate of 0.234; ECE 0.028); the reaction to a terrorist death overshoots by about five points (0.728 against 0.678; ECE 0.052). Fig. 17 shows four example rounds and the aggregate answer to whether the curve favours the stronger team at freeze-end. At equal buys the higher-ranked team wins 53–59 % of rounds when the HLTV rank gap is six or more (pistol rounds and even full-buy rounds), but the production model’s first value is 0.50–0.51 for the favourite, because it has no team features; the stack with same-year priors tilts to 0.53–0.56, the honest lagged stack to 0.51–0.52. A five-point favourite at equal money is the size of the pre-round strength effect in this game, and it is invisible in the deployed curve.

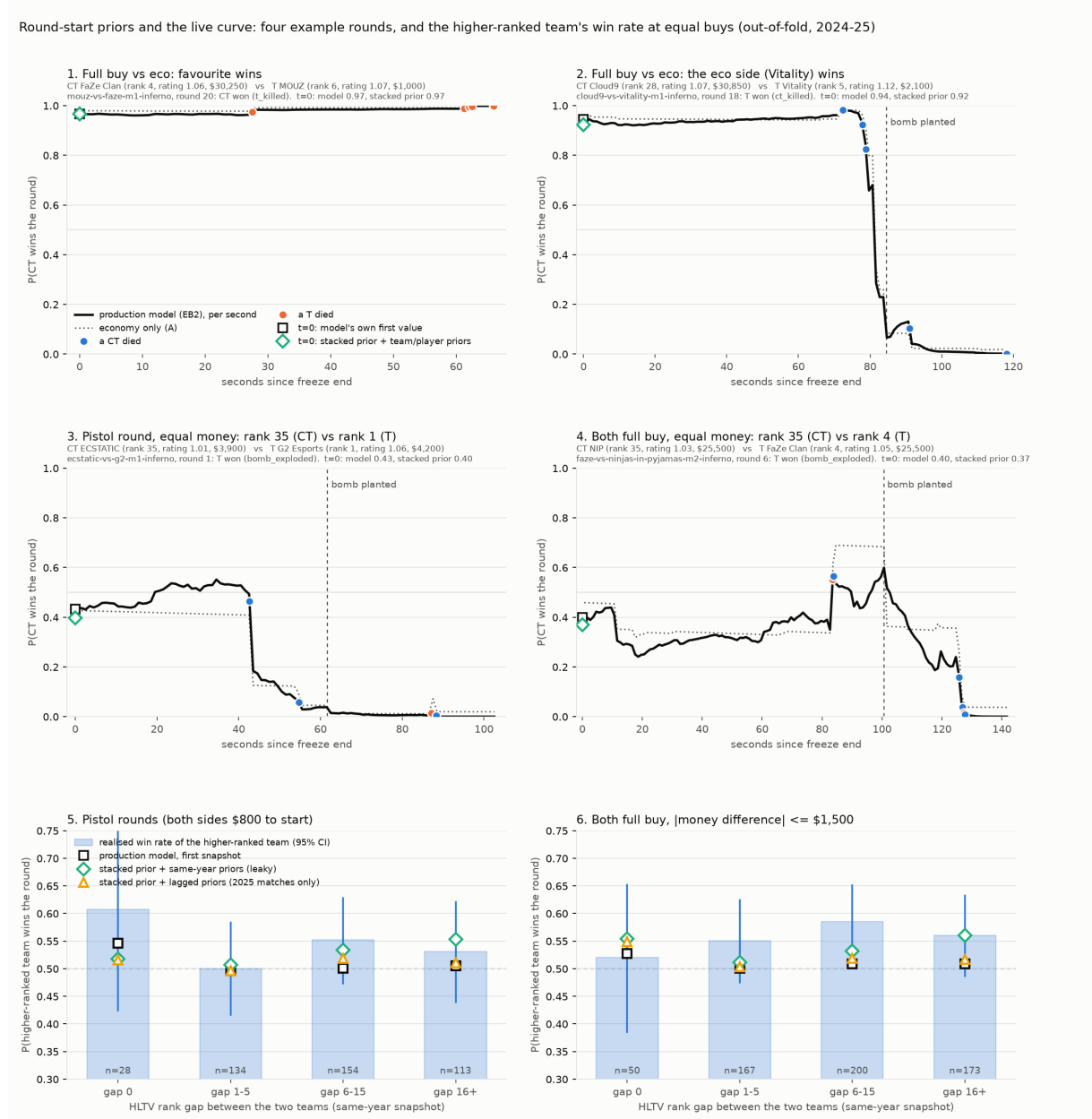


Figure 17: Round-start priors and the live curve. Panels 1–4: per-second $P(\text{CT win})$ from the production model (out-of-fold) for a full-buy versus eco round won by the favourite, one won by the eco side, a pistol round between the 35th- and 1st-ranked teams, and an equal full-buy round between the 35th- and 4th-ranked teams; dots mark deaths (blue: a CT died, orange: a T died), the dashed line the bomb plant, and the markers at $t = 0$ the model’s own first value (square) and the stacked prior with team and player priors (diamond). Panels 5–6: realised win rate of the higher-ranked team at equal buys by rank gap, with 95 % intervals, against the round-start models.

7.13 Per-second honesty during a defuse

The trajectory diagnostic above is global; here is a concrete, local instance of the same idea. Every bomb feature in Sect. 7.7 is *counterfactual*: the defuse-race margin and its relatives ask, from geometry, whether a CT *could* reach the bomb and finish in time. None of them observes a defuse that is *actually happening*. The cost is a flat curve: on the 2026 test set, as a defuse runs from start to completion the empirical CT win rate climbs from 0.91 to 0.98, while the EB2 model, carrying the entire counterfactual bomb block, barely moves off 0.92 (Fig. 18).

We add four columns that measure the live defuse: whether one is in progress, seconds elapsed, the kit-aware fraction of the required time completed, and whether the remaining defuse fits inside the remaining fuse. They are derived from the per-tick defusing flag before down-sampling, which is why the demos were re-parsed. They extend the bomb-geometry pillar (Appendix C) rather than forming a pillar of their own, but because a near-complete defuse is close to an endgame certainty – crediting it to the spatial pillars would inflate them – we keep them out of the headline sets and toggle them as an ablation (EB2D, EFB2D denote EB2 and EFB2 with these four columns added). The feature is not a relabelling of the outcome: 39.6 % of the 1,032 defuse attempts in training are interrupted, and the model stays calibrated on those.

Because the feature fires on only 0.97 % of snapshots, it cannot and does not move a pooled metric: EB2D and EB2 are indistinguishable on overall AUC and on contested-AUC. Its value is entirely on the defusing rows, and there it is large. Adding it cuts log-loss on those rows by **60 %** ($0.158 \rightarrow 0.063$) and Brier by 62 % for the logistic model, and the predicted curve tracks the empirical one almost exactly (Fig. 18). This is the contested-AUC argument in miniature: a metric pooled over all snapshots hides a six-point calibration error at the single most consequential moment a live overlay displays, and a slice reveals it.

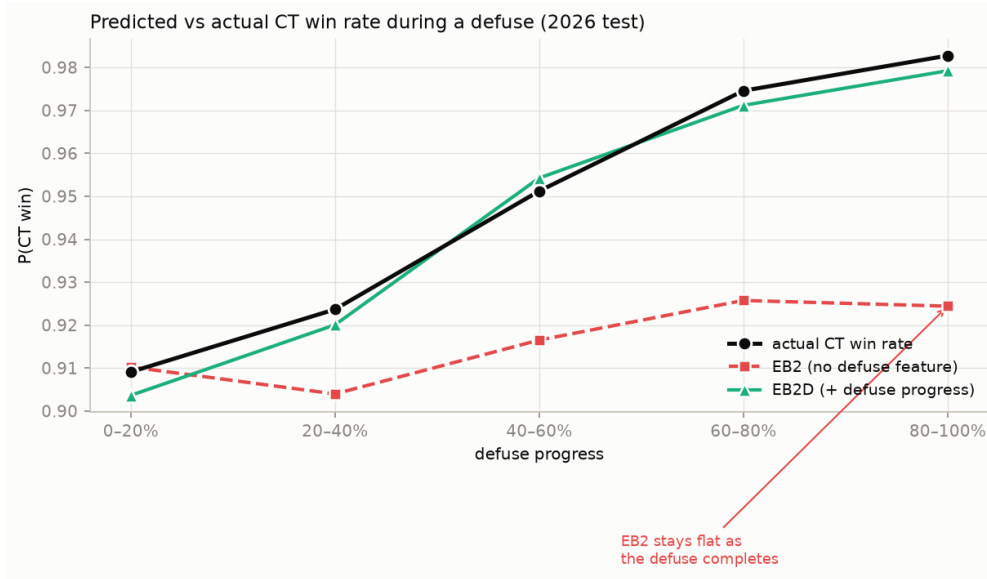


Figure 18: Predicted versus actual CT win rate as a defuse progresses (2026 test, logistic). The EB2 model (red) stays flat as the defuse completes; adding the four defuse-progress columns (EB2D, green) recovers the empirical curve (black). The feature touches $\sim 1\%$ of snapshots, so it is invisible in pooled AUC while removing most of the calibration error on the defusing rows.

The effect is not specific to the logistic model. The two sequence models, trained on all 2024–25 data and evaluated once on the 2026 holdout, consume every feature column—the four defuse-progress columns included—and their pooled out-of-time AUC is statistically indistinguishable from the classical models (TCN 0.843, Transformer 0.838; every 95% interval overlaps the clas-

sical ~ 0.85), so the dead heat of Sect. 7.10 persists here too. On the 631 defusing rows the calibration correction carries over out-of-time (Table 8): both sequence models track the empirical curve and sit far below the featureless baseline. The best-calibrated model on this slice is nonetheless the logistic EB2D, ahead of both networks that also ingest the feature—a concrete instance of the paper’s recurring finding that the physics-derived signal, not model capacity, is what the pooled metric was missing.

Table 8: Calibration on the 631 defusing rows of the 2026 holdout. Every row except the first includes the four defuse-progress columns. The logistic model with the feature is the best calibrated, ahead of the sequence models that also consume it.

Model (defusing rows)	log-loss	Brier
Logistic, EB2 (no feature)	0.158	0.045
Logistic, EB2D (+ feature)	0.063	0.017
TCN (+ feature)	0.083	0.027
Transformer (+ feature)	0.124	0.038

7.14 What the model looks like live

Figure 19 is the actual deliverable: a per-second win-probability curve with a bootstrap uncertainty band, kills and the bomb plant marked.

It repays study, because it shows both the model’s competence and its limits. The curve is *step-like*: it is essentially flat between kills and jumps at them. The model is, to a first approximation, a kill-counter with a good sense of what each kill is worth in context, which is a fair description of what economy-plus-state features can be. And the uncertainty band is *wide* (mean width 0.21): the model knows it does not know. Note the plant at $t = 55$ s pushes $P(\text{CT win})$ down to 0.25 and holds it there, flat, for twenty seconds, the defuse-race feature keeping the CTs written off until a late kill re-opens the round.

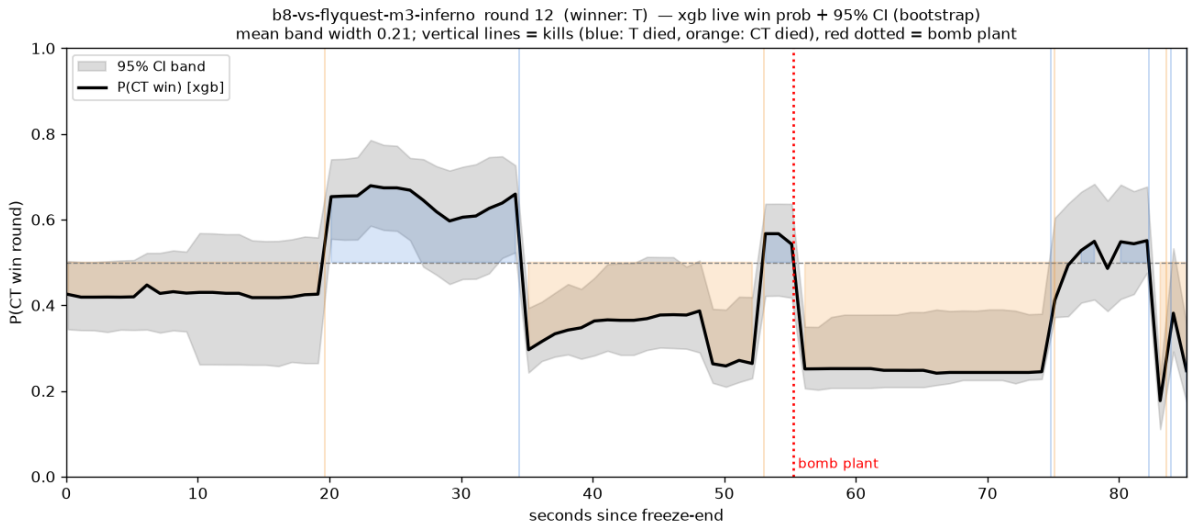


Figure 19: A live win-probability curve with 95% bootstrap band. Vertical lines are kills (blue: a T died; orange: a CT died); the red dotted line is the bomb plant. The curve is step-like (flat between kills, jumping at them) and the band is wide (mean 0.21). Both are honest reflections of what this model does and does not know.

7.15 The out-of-time holdout: the model transfers, its skill prior does not

We trained each model on the full 220-match set (2024–25) and evaluated **once** on the 27 unseen 2026 matches.

Table 9: In-time (out-of-fold) vs. out-of-time (2026) AUC.

Set	Model	In-time	Out-of-time	Δ
EB2	LightGBM	0.8493	0.8501	+0.0008
EB2	XGBoost	0.8489	0.8497	+0.0007
E	XGBoost	0.8476	0.8476	0.0000
EFB2	Logistic reg.	0.8519	0.8236	−0.0283
EFB2	Random forest	0.8446	0.8159	−0.0287
EFB2	LightGBM	,	,	−0.0155
EFB2	XGBoost	,	,	−0.0136

Figure 20 is the result in one picture. Points on the diagonal did not degrade. **Every non-firepower configuration (blue, orange, green) sits on or above the line. Every firepower configuration (red diamonds) falls below it**, and the further a firepower model was ahead in-sample, the further it falls.

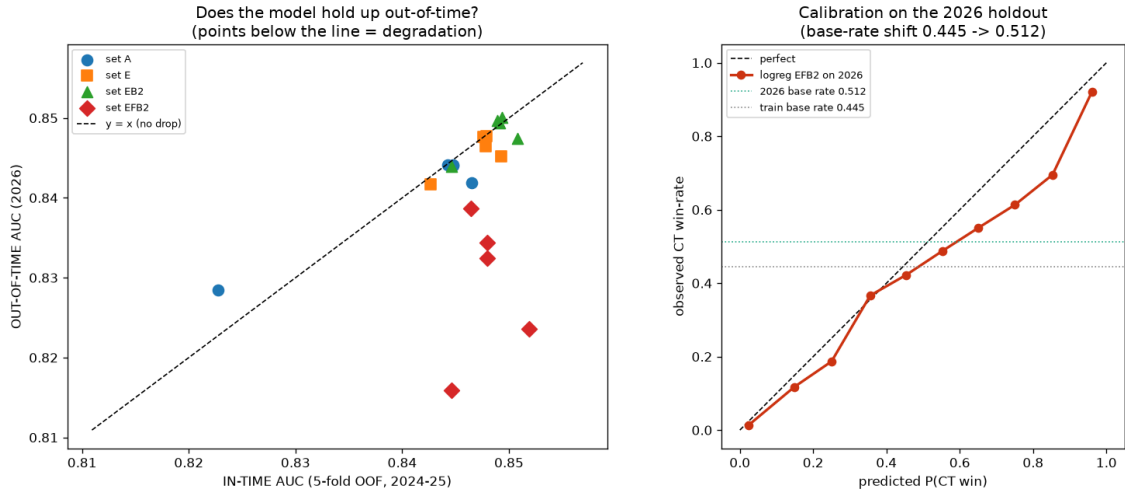


Figure 20: Left: in-time vs. out-of-time AUC. Points below the dashed line degraded. Sets A/E/EB2 (no firepower) sit on the line, they transfer. Every EFB2 point (red diamonds, with firepower) falls well below it. **Right:** calibration of the best in-sample model (logreg EFB2) on the holdout. It now sits systematically *below* the diagonal: over-confident in the CTs across the whole range (intercept -0.36), even though the true 2026 base rate (0.512) is *higher* than training (0.445).

The demo-derived pillars transfer essentially perfectly. Economy, Voronoi map control, tactical readiness and defuse-race geometry all generalise to a new season, new game patches and substantially new rosters. Contested-AUC even *improves* out-of-time (0.590 $\rightarrow$ 0.64–0.65). The non-firepower sets show only a benign base-rate drift (calibration slope ≈ 1.0 , intercept $+0.11$ to $+0.13$), the expected consequence of the $0.445 \rightarrow 0.512$ shift in CT win rate, and trivially correctable with a single intercept adjustment.

But the best in-sample model becomes the worst out-of-time. Logistic EFB2 (the study’s best cross-validated model) loses 0.0283 AUC. Calibration breaks with it: ECE 0.016 $\rightarrow$

0.071, calibration intercept -0.36 . The right panel of Fig. 20 shows the damage: the reliability curve is dragged below the diagonal across the entire range. The probabilities become untrustworthy, which for a live product is the more serious of the two failures, a model that is merely less accurate degrades gracefully, whereas a model that is confidently wrong misinforms.

Diagnosis: a data-coverage failure, not a signal failure. Figure 21 is the diagnosis. In 2026, $\approx 30\%$ of players have no entry in the HLTV skill database, rookies, stand-ins, and players who have not yet accumulated a season of Tier-1 statistics, and their contribution to the team skill-sum silently becomes zero. The feature, tightly clustered at a mean of 5.28 at 5-v-5 in training (all five players known), collapses to a mean of 3.66 on the holdout with a long tail toward zero: **11.8 % of 2026 5-v-5 snapshots have a team skill-sum below 3, versus 0.0 % in training.**

The same feature means something different in 2026. And (this is the cruel part) because it is a *sum over alive players* (Sect. 7.8), a degraded value is indistinguishable from a legitimate one: “skill-sum 3” means *three good players alive* in training and can mean *five players, two of whom are unknown* in 2026. The model reads the corrupted input as a real, familiar game state and mispredicts with confidence. That is why the failure is silent, and why it damages calibration rather than merely accuracy.

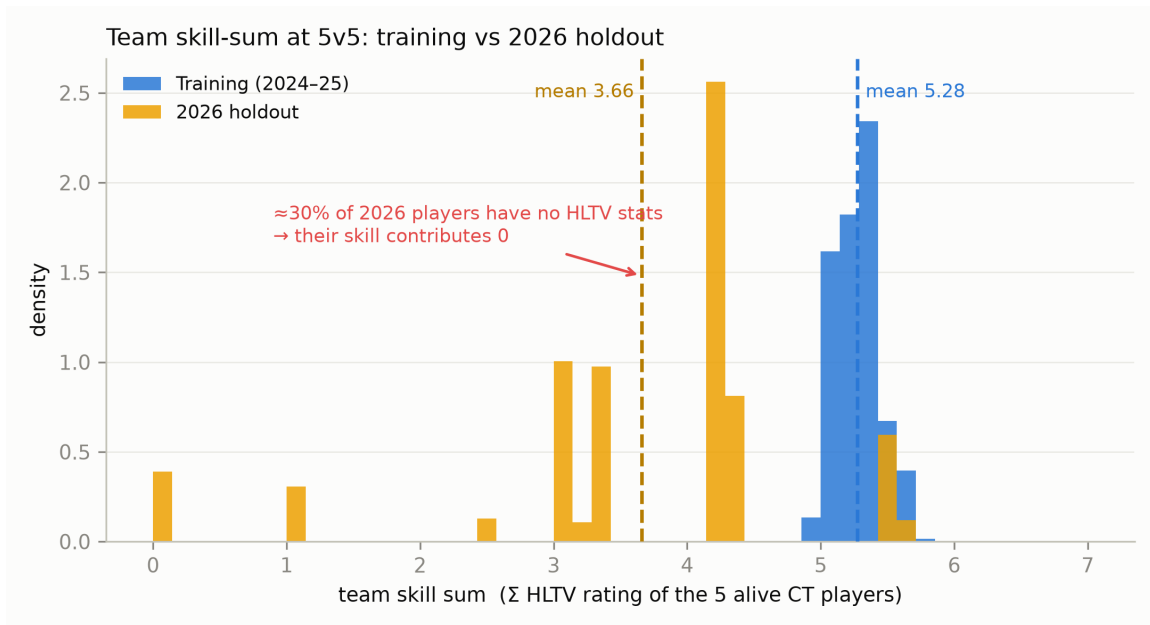


Figure 21: Why firepower fails out-of-time. The team skill-sum at 5-v-5, tightly clustered in training (mean 5.28; all five players known), collapses on the 2026 holdout (mean 3.66) because $\approx 30\%$ of players have no database entry and silently contribute zero.

The lesson. Pillars 1–3 are computed *from the observation itself* and are therefore always available at inference time. Pillar 4 is a *prior sourced from outside the observation*, and therefore carries an **inference-time data dependency**: at deployment it needs an external database, indexed by an era which (by definition) is the era you have least data for. When that database lags, the feature does not merely stop helping. **It actively harms the model, and it does so silently.**

Cross-validation cannot see this. Every fold is drawn from the same era, in which the database is complete; the feature is fully populated in both halves of every split. *The failure is invisible by construction to the standard protocol.* We therefore argue that any model incorporating

externally-sourced priors requires an out-of-time gate, and that its production configuration should be chosen on out-of-time evidence rather than cross-validated evidence.

Ours is EB2 (with no firepower at all) even though cross-validation ranked EFB2 first.

7.16 Three data constructions: fixing the data does not rescue the pillar

The failure above is a *coverage* failure, so the obvious question is whether it survives a correct construction. We answer it exhaustively, with **three** constructions of the skill prior that trace out a spectrum of data quality (Fig. 22):

1. **Same-year, broken** (Sect. 7.15): 2026 matches, no 2026 statistics existed, $\approx 30\%$ zero-filled. The coverage-gap catastrophe.
2. **Lagged-2025**: each 2026 match looks up the previous season’s skill, the *leak-free*, deployment-realistic prior (a 2025 rating cannot be computed from a 2026 match; it is exactly what a live system would have at match time). 100% coverage.
3. **Same-year 2026**: each 2026 match looks up that season’s statistics, 100% coverage *and* same-era knowledge, but *leaky* (a 2026 rating is computed partly from the very matches being predicted, and would not exist live). This is the *best case*: the most information a skill prior could possibly carry.

We collected both the 2025 and 2026 statistics for all 82 players (100% coverage each) and re-ran the classical holdout for each variant as **separate, disclosed** evaluations; the touch-once run of Sect. 7.15 stands unchanged. In both corrected variants the coverage gap closes: the 5-v-5 team skill-sum, which had collapsed to a mean of 3.66 on the broken build, returns to 5.36 (lagged) / 5.42 (same-year), against a training mean of 5.28, with 0.0% of snapshots below 3 (was 11.8%).

The result separates cleanly into two findings, one about calibration and one about value.

Coverage is the entire calibration story. The smoking gun of Sect. 7.15, a calibration intercept of -0.36 and ECE 0.071 , appears *only* in the broken build. Both coverage-complete variants sit at the benign near-zero intercept the demo-only sets show, with ECE back in the healthy 0.02 – 0.03 band (Fig. 22, right). Whether the statistics are leak-free (lagged) or leaky same-era makes no difference to calibration: the miscalibration was *entirely* a coverage artifact.

Firepower earns its place under no construction. Discrimination recovers most of the collapse in both corrected variants, but neither overtakes the demo-only model. **EFB2 remains below EB2 on every single model in every variant, including the leaky best case** (Table 10). Same-year 2026 (full coverage *and* an oracle’s same-season knowledge) still trails EB2 by 0.002 to 0.011 AUC on all five models, and its contested-AUC is worse (0.55 – 0.59 vs EB2’s 0.63 – 0.66) in every variant. Notably the lagged and same-year builds are nearly identical: *which season* of statistics we sum barely matters, exactly as the count confound of Sect. 7.8 predicts for the summed encoding these sets carry ($r = 0.987$ with the player count, so the feature mostly counts players and the ratings inside the sum are almost immaterial). The in-sample edge of firepower (logistic EFB2 0.8519 , the study’s best in-sample number) is an in-sample artifact that reverses out-of-time under every construction.

Two lessons, cleanly separated by the three variants. The spectrum isolates two distinct properties of an external prior. (i) *Plumbing*. An inference-time data dependency can fail catastrophically and *silently* when coverage lags, and that failure is fixable, and is purely about coverage (the calibration disaster vanishes the moment the database is complete, regardless

Table 10: EFB2 out-of-time AUC on the 2026 holdout under all three skill-prior constructions, against the demo-only EB2. Firepower is a net negative on every model in every variant, including the leaky best case (same-year 2026, full coverage + same-era knowledge). Bold = best out-of-time per row.

Model	EB2 (no fp)	broken	lagged-2025	same-year 2026
Logistic	0.8474	0.8236	0.8423	0.8450
XGBoost	0.8497	0.8344	0.8401	0.8399
LightGBM	0.8501	0.8325	0.8393	0.8386
CatBoost	0.8494	0.8386	0.8377	0.8391
Random forest	0.8439	0.8159	0.8378	0.8422

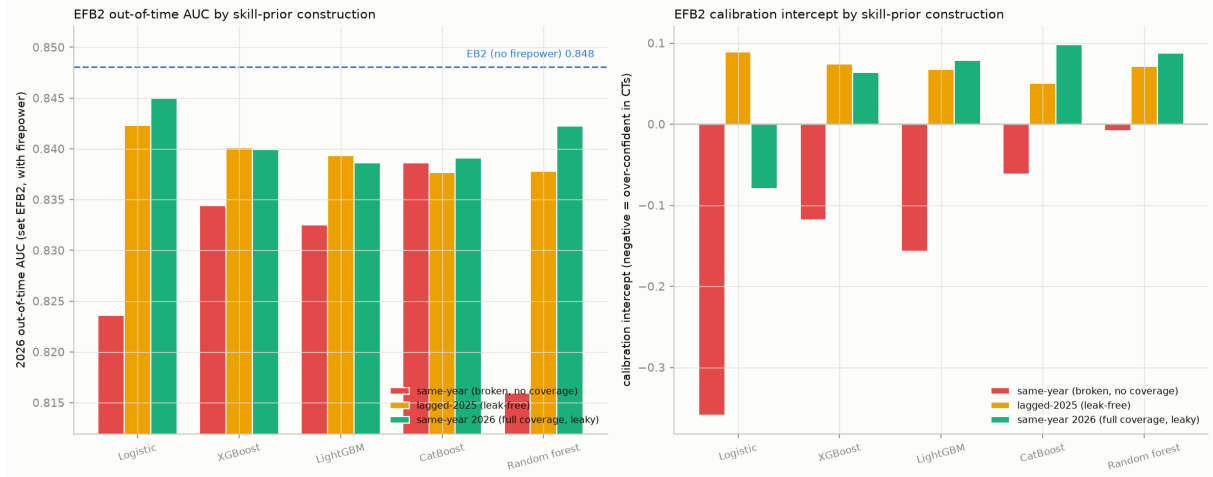


Figure 22: Firepower under three data constructions, EFB2 on the 2026 holdout. **Left:** out-of-time AUC per model; every firepower bar (broken (red), leak-free lagged (yellow), and the leaky best-case same-year (green)) sits below the EB2 no-firepower line (blue dashed). No construction of the skill prior beats the demo-only model. **Right:** calibration intercept; only the broken build is miscalibrated (over-confident in the CTs), while both coverage-complete builds sit near zero. The y -axis on the left is truncated to $[0.81, 0.85]$ to resolve the differences; all bars are $\text{AUC} \approx 0.82\text{--}0.85$.

of which season it draws from). (ii) *Value*. Even correctly plumbed (and even given a leaky oracle’s same-season knowledge that no live system could have) the skill prior adds no out-of-time value here: the demo-derived pillars already capture what matters, and EB2 remains the best configuration. A reviewer seeing only the broken run might reasonably say “fix the data”; a reviewer seeing only the lagged run might say “the lag costs you the signal.” We ran all three, disclosed them, and firepower transfers under none. That is the strongest form of the claim. The production recommendation of Sect. 7.15 therefore stands, definitively: **ship EB2, a correctly-constructed, coverage-complete, even leaky skill prior does not beat the demo-only model out-of-time.**

We note one mild caveat common to all three variants: the EFB2 models are *trained* on same-year firepower and *tested* on the holdout prior. For the lagged variant this is the realistic deployment setup (train on history, serve with last season); for the same-year variant train and test eras are consistent. The 2026 distributions match training closely, so the train/serve skew is small.

8 Discussion

Report contested-AUC. Pooled AUC on a per-second win-probability model is close to a measure of how often the game is lopsided. Future Counter-Strike win-probability work should

report performance on the even-round subset, where all current models sit near chance and where improvement actually matters. We would go further: a model that improved contested-AUC from 0.58 to 0.62 while *losing* a point of pooled AUC would be the more valuable model, and the current convention would reject it.

Two forms of honesty. The contested-AUC and trajectory-calibration results are the same lesson applied to two different over-readings of a win-probability feed. Pooled AUC over-states how much a model *knows* in even rounds; a naive reading of a probability trajectory over-states how *surprising* its comebacks are. Reporting contested-AUC guards against the first, and benchmarking the path extremes against the martingale law guards against the second. A trustworthy live model should be humble on both axes: it should not claim discrimination it lacks in balanced rounds, and it should not sensationalise the swings it does show. Our recommended configuration is honest on both.

Stability beats fidelity. The grey-vs-territory result suggests a general principle for tracking data: when the target is a *future* outcome, a representation’s *temporal stability* can matter more than its instantaneous correctness. Practitioners building vision- or reachability-based control surfaces should test a memory-augmented variant before concluding that control “does not help”.

Physically-derived features beat learned ones at this scale. The defuse-race margin (three quantities and a subtraction) outperformed everything the deep models discovered on their own from the same underlying distances. Figures 2 and 9 are the same geometry under two encodings; only one makes the feasibility boundary learnable. With $\approx 5k$ rounds, encoding known physics is a better use of capacity than learning it.

Beware the external prior. This is the finding we most want to travel. A “skill prior”, “player form”, “team strength” or any feature joined from an outside table introduces a dependency that cross-validation is structurally blind to. Such a feature should be (a) gated by an out-of-time evaluation, (b) constructed as a *lag*, using only information that existed before the event being predicted, and (c) monitored for *coverage* in production, with an explicit fallback (a per-capita encoding, an imputed league average, or dropping the pillar) for when coverage drops. None of these is exotic. What is easy to miss is that *nothing in the standard protocol tells you that you need them*.

9 Limitations

1. **Single map.** All results are *de_inferno*. Spatial features are map-specific by construction; multi-map generalisation is untested.
2. **Scale.** 220 matches, $\approx 4,900$ rounds. The statistical equivalence of the deep and classical models is a statement about this regime and should not be read as a general claim.
3. **Firepower leakage, and the train/serve skew of the fix.** Same-era player ratings are partly computed from the matches being predicted; the lagged-prior variant (Sect. 7.16) is the clean construction and is what we recommend acting on. Its one caveat is that the models are trained on same-year firepower and evaluated on the lagged prior, a small, deployment-realistic skew.
4. **Firepower count confound, repaired.** The summed-rating feature is approximately a player-count proxy ($r = 0.987$; Fig. 10). Sect. 7.8 repairs it by averaging rather than summing, and tests two further encodings built on the repaired base. Removing the

confound helps, and not enough: out-of-time the mean encoding still sits behind the skill-free model. The sets used elsewhere in the paper retain the summed encoding, which is not the strongest of the four; the out-of-time conclusion was re-checked with the cross-validation-preferred encoding and is unchanged.

5. **Irreducible ceiling.** Contested rounds sit near 0.58 for *every* model tested. Some of this is genuine aleatoric randomness, a 5-v-5 on equal buys really is close to a coin flip, and it should be. How much is irreducible and how much is headroom is unknown, and is the most important open question this work leaves.
6. **Deep models consume all features, so they carry firepower.** The TCN and Transformer were evaluated out-of-time (Sect. 7.10); they tie the classical models there too. Because they ingest every feature column, they inherit the firepower penalty and cannot be run in a “no-firepower” configuration without retraining on a reduced column set, which we did not do. A deep model restricted to the demo-derived pillars might close the small gap to EB2; we leave that to future work.
7. **Single holdout era.** One out-of-time window (2026). The failure mode we identify is general, but its *magnitude* here is a single observation.
8. **Discrete, jump-driven trajectories.** The trajectory-calibration benchmark (Sect. 7.11) is exact only for continuous forecast paths. Counter-Strike win probability moves in discrete jumps at kills, so the observed write-off frequencies are compared against a conservative continuous reference rather than the exact discrete law. This is sufficient to establish the absence of over-reaction, but distinguishing the residual conservatism of a discrete forecast from mild under-reaction of a per-snapshot model would require the discrete correction, which we leave to future work.
9. **Round-start prior is in-time only.** Sect. 7.12 is evaluated by cross-validation on the 220 training matches and has not been scored on the 2026 holdout. Its lagged priors are undefined for the 2024 matches because no 2023 player or team tables exist, so the lagged rungs rest on the 2025 matches alone.

10 Conclusion

We built a per-second win-probability model for Counter-Strike 2 and used it to interrogate three conventions of the field.

Pooled AUC flatters: restricted to even rounds, every model is near a coin flip, and we propose contested-AUC as the honest metric. Physical fidelity does not imply predictiveness: the most realistic map-control model is the least useful until it is given memory. And cross-validation is not a deployment test: our best cross-validated model became our worst out-of-time, because one pillar depended on a database the future does not yet have, a failure mode that grouped folds cannot expose. And when we repaired that dependency across three constructions (a leak-free lagged prior and even a leaky same-era oracle) the pillar’s calibration recovered fully but it *still* did not beat the demo-only model under any of them, so the negative verdict is not about broken data but about the value of external priors themselves.

The features that *did* transfer (economy, Voronoi map control, and a three-term defuse-race inequality) are the ones computed directly from the game itself.

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A Reproducibility notes

Parsing. awpy v2.0.2, tickrate forced to 64.

Splits. 5-fold GroupKFold keyed on `match_id`. The 2026 holdout lives in a physically separate directory tree and is assembled through a separate code path, so no glob over the training directories can sweep it in.

Bootstrap. $B = 500$ resamples of *matches* (with replacement), recomputing all metrics on fixed out-of-fold predictions. For the holdout, the 27 test matches are the resampling unit.

Compute. Classical models train on a laptop CPU in minutes. The three deep models were trained on a single NVIDIA B200 GPU (PARCC Betty). Each was trained for up to 40 epochs with early stopping (patience 6) on an inner validation split, which in practice halted near epoch 15. At roughly 0.3–0.4s per epoch, a full 5-fold cross-validation completes in a few minutes of GPU time per model, so the deep-model cost is modest despite the GPU requirement.

B Full metric battery

Table 11 reports the in-time (5-fold out-of-fold) metric battery for every classical model and feature set, together with the deep models and the ensemble. Table 12 reports the out-of-time (2026 same-year holdout) battery for the two feature sets that matter for the deployment recommendation, EB2 and EFB2, including calibration slope and intercept. Metric definitions are given in Sect. 6; all match-level bootstrap confidence intervals for the headline comparisons appear with the corresponding figures in the main text.

Table 11: In-time metric battery (5-fold GroupKFold out-of-fold). AUC, log-loss, Brier, expected calibration error (ECE), Brier Skill Score (BSS), and contested-AUC. The deep and ensemble rows use the all-pillar feature set.

Set	Model	AUC	log-loss	Brier	ECE	BSS	cont.-AUC
A	Logistic	0.8465	0.4675	0.1579	0.019	0.361	0.587
A	XGBoost	0.8443	0.4687	0.1588	0.018	0.357	0.580
A	LightGBM	0.8448	0.4680	0.1585	0.017	0.358	0.582
A	CatBoost	0.8448	0.4666	0.1582	0.016	0.359	0.580
A	Random forest	0.8228	0.5305	0.1718	0.042	0.305	0.543
E	Logistic	0.8493	0.4605	0.1564	0.016	0.367	0.590
E	XGBoost	0.8476	0.4618	0.1569	0.012	0.365	0.585
E	LightGBM	0.8479	0.4616	0.1568	0.014	0.365	0.586
E	CatBoost	0.8478	0.4616	0.1567	0.011	0.366	0.590
E	Random forest	0.8426	0.4747	0.1596	0.011	0.354	0.585
EB2	Logistic	0.8508	0.4575	0.1556	0.015	0.370	0.596
EB2	XGBoost	0.8489	0.4586	0.1561	0.014	0.368	0.586
EB2	LightGBM	0.8493	0.4585	0.1560	0.013	0.369	0.590
EB2	CatBoost	0.8491	0.4590	0.1560	0.013	0.368	0.588
EB2	Random forest	0.8446	0.4684	0.1585	0.010	0.358	0.589
EFB2	Logistic	0.8519	0.4554	0.1550	0.016	0.373	0.603
EFB2	XGBoost	0.8480	0.4601	0.1566	0.019	0.366	0.589
EFB2	LightGBM	0.8479	0.4602	0.1566	0.018	0.366	0.589
EFB2	CatBoost	0.8465	0.4627	0.1573	0.015	0.363	0.584
EFB2	Random forest	0.8446	0.4661	0.1581	0.014	0.360	0.579
EFB2	TCN	0.8489	0.4588	0.1560	0.012	0.368	0.572
EFB2	Transformer	0.8473	0.4612	0.1568	0.009	0.365	0.568
EFB2	Ensemble	0.8531	0.4520	0.1537	0.009	0.378	0.589

Table 13: No-economy ablation, both models: AUC in-time (5-fold out-of-fold) and on the single 2026 scoring. Sets as in Table 1.

Feature set	Logistic	Logistic 2026	XGBoost	XGBoost 2026
EB2, production set	0.851	0.847	0.850	0.850
A, economy block	0.847	0.842	0.845	0.844
Everything except money	0.841	0.841	0.844	0.844
Money only	0.830	0.823	0.834	0.829
All non-economy pillars	0.822	0.822	0.830	0.829
Voronoi + tactical	0.818	0.815	0.828	0.825
All non-economy, no count columns	0.806	0.802	0.823	0.820
Voronoi + tactical, no count columns	0.801	0.793	0.820	0.815
Combat state only	0.797	0.787	0.800	0.799
Voronoi control only	0.723	0.703	0.745	0.732
Bomb geometry + defuse race	0.686	0.675	0.695	0.671
Round clock only	0.555	0.539	0.550	0.539
Score and round number	0.536	0.527	0.537	0.558

C Complete feature list

Table 15 enumerates every feature by pillar, with a short description. Symmetric counterpart features for the two sides are written once as `{ct,t}_...` and counted as two. The feature sets used in the main text are compositions of these pillars:

Table 12: Out-of-time metric battery on the 2026 same-year holdout, for EB2 (no firepower) and EFB2 (with firepower), including calibration slope and intercept. EB2 dominates EFB2 on every model on AUC, log-loss, Brier, and contested-AUC.

Set	Model	AUC	log-loss	Brier	ECE	cont.-AUC	cal. slope	cal. int.
EB2	Logistic	0.8474	0.4663	0.1583	0.029	0.655	0.99	+0.10
EB2	XGBoost	0.8497	0.4632	0.1571	0.028	0.634	1.01	+0.12
EB2	LightGBM	0.8501	0.4625	0.1568	0.023	0.637	1.02	+0.12
EB2	CatBoost	0.8494	0.4637	0.1572	0.025	0.654	1.06	+0.12
EB2	Random forest	0.8439	0.4825	0.1606	0.022	0.580	0.97	+0.12
EFB2	Logistic	0.8450	0.4687	0.1590	0.022	0.585	0.94	−0.08
EFB2	XGBoost	0.8399	0.4742	0.1614	0.030	0.574	0.95	+0.06
EFB2	LightGBM	0.8386	0.4747	0.1617	0.030	0.572	0.95	+0.08
EFB2	CatBoost	0.8391	0.4754	0.1618	0.027	0.593	1.00	+0.10
EFB2	Random forest	0.8422	0.4813	0.1614	0.020	0.553	1.12	+0.09

Table 14: Feature-set composition. Each set is the union of the pillar blocks marked.

Set	Pillars included	Count
A	Economy & combat	17
E	A + Voronoi map control + Tactical + bomb-plant geometry	60
EB2	E + bomb-live + defuse-race	72
EFB2	EB2 + territory map control + Firepower	97

The four live-defuse columns belong to the bomb-geometry pillar (Table 15) and are not a separate set. They are evaluated as an ablation in Sect. 7.13: EB2D and EFB2D denote EB2 and EFB2 with those four columns added (76 and 101 features respectively).

Table 15: Complete feature inventory, grouped by pillar. {ct,t} denotes a symmetric pair, one feature per side.

Feature	Description
Pillar 1 Economy & combat state (17)	
time_elapsed_sec	Seconds since freeze-end.
{ct,t}_players_alive	Living players on each side.
{ct,t}_health_total	Summed health of living players.
{ct,t}_armor_total	Summed armour of living players.
{ct,t}_equipment_value	Summed dollar value of held equipment.
{ct,t}_economy_class	Buy tier (eco / force / full) per side.
ct_defuse_kits	Number of defuse kits held by CTs.
bomb_planted	Whether the bomb is currently planted.
{ct,t}_score	Rounds won so far this half.
score_diff	CT score minus T score.
round_num	Round index within the match.
Pillar 2 Map control (24)	
ct_voronoi_control_pct	CT share of the map under Voronoi (nearest-player) control.
control_deficit	CT minus T controlled area.
ct_fa_site,b_site,banana,mid,ct_splavot,ctfract	CT share of each macro-zone (5).
control_trend	Short rolling change in CT control.
control_volatility	Rolling variance of CT control.
{ct,t}_los_control	Instantaneous LOS+FOV+smoke control per side (grey model).

Table 15 continued

Feature	Description
contested_pct	Map share contested by both sides (grey model).
grey_pct	Map share controlled by nobody (grey model).
ct_los_deficit	CT minus T instantaneous control.
{ct,t}_terr_control	Territory-model control per side (grey + 15s memory).
terr_contested_pct	Contested share, territory model.
terr_grey_pct	Neutral share, territory model.
ct_terr_deficit	CT minus T territory control.
terr_{a_site,b_site,banana,mid,ct_spawn_t,defuse_t}	Respawn/defuse deficit (5).
Pillar 3 Tactical readiness & bomb geometry (54)	
{ct,t}_positional_entropy	Shannon entropy of players over zones (spread).
{ct,t}_{a_site,b_site,banana,mid,ct_spawn_t,defuse_t}	Player spawn/defuse per side (10).
{ct,t}_awp_alive	Whether a living player holds an AWP.
{ct,t}_awp_zone	Zone of the AWP holder.
bomb_carrier_zone	Zone of the bomb carrier (pre-plant).
{ct,t}_{smokes,flashes,fire,he}	Grenade counts by type per side (8).
{ct,t}_util_total	Total grenades held per side.
utility_advantage	CT minus T total utility.
bomb_site	Plant site (A / B / none).
bomb_plant_{x,y}	Plant coordinates.
min_ct_dist_to_bomb	Nearest-CT straight-line distance to the bomb.
min_ct_path_to_bomb	Nearest-CT nav-mesh path distance to the bomb.
n_ct_near_bomb	CTs within a fixed radius of the bomb.
bomb_state	Carried / dropped / planted.
bomb_dropped	Whether the bomb is loose on the ground.
ct_bomb_local_control	CT control of the area around the live bomb.
ct_bomb_local_deficit	CT minus T local bomb-area control.
min_{ct,t}_dist_to_bomb_live	Nearest per side to the live bomb.
ct_closer_to_bomb	Whether a CT is nearer than any T.
defuse_time_margin	Fuse time – (nearest-CT path time + defuse duration).
defuse_margin_kit	Per-CT kit-aware defuse margin.
n_ct_can_defuse	CTs who could physically finish a defuse in time.
best_defuser_has_kit	Whether the fastest potential defuser holds a kit.
defuse_contest_margin	Defuse margin net of the nearest T’s interrupt window.
<i>live-defuse tracking (4) the four features above are counterfactual (could a CT defuse in time); these four measure an actual running defuse, added in the re-parse (Sect. 7.13), and toggled as an ablation (EB2D / EFB2D)</i>	
defuse_in_progress	Whether a defuse is being performed right now.
defuse_elapsed_sec	Seconds into the current defuse attempt.
defuse_progress_frac	Elapsed / required (kit-aware), in [0, 1].
defuse_beats_fuse	Whether the remaining defuse fits inside the remaining fuse.
Pillar 4 Firepower / skill prior (20, v2 sums)	
{ct,t}_rating_sum	Summed HLTV rating of living players.
{ct,t}_adr_sum	Summed average damage per round.
{ct,t}_kast_mean	Mean KAST of living players.
{ct,t}_hltn_firepower_sum	Summed HLTV firepower rating.
{ct,t}_entry_sum	Summed entry-fragging rating (teammates alive).
{ct,t}_trading_sum	Summed trading rating (teammates alive).
{ct,t}_opening_sum	Summed opening-duel rating (5-v-5 only).
{ct,t}_clutch_score	Lone-survivor clutch rating.
{ct,t}_awp_sniping_skill	Sniping rating of the AWP holder.
{ct,t}_weighted_utility	Utility skill weighted by grenade value held.
Interactions (4)	

Table 15 continued

Feature	Description
<code>ctrl_x_eveneco</code>	Voronoi control $\times$ even-economy indicator.
<code>terr_x_eveneco</code>	Territory control $\times$ even-economy indicator.
<code>ctrl_x_equalalive</code>	Voronoi control $\times$ equal-players-alive indicator.
<code>terr_x_equalalive</code>	Territory control $\times$ equal-players-alive indicator.

D Supplementary figures

We include two model-interpretation figures that support the main-text claims but are not essential to follow the argument.

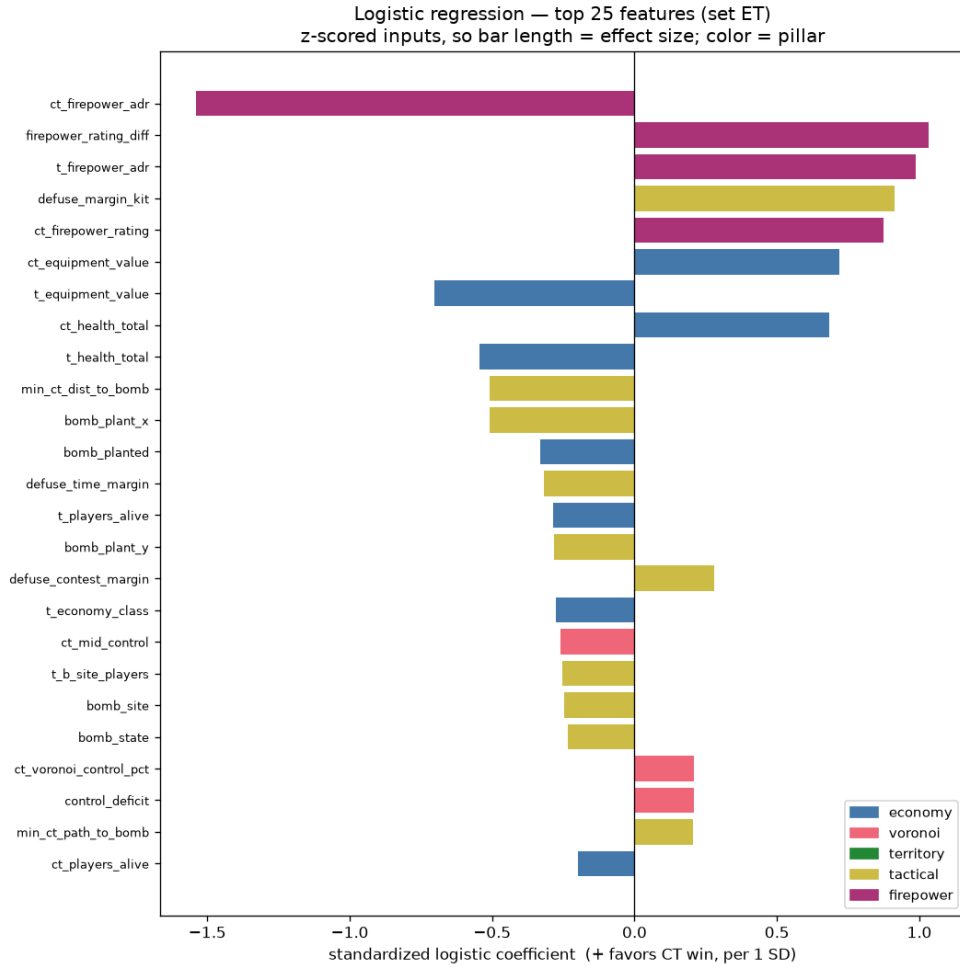


Figure 23: Standardised coefficients of the logistic regression (top 25 features, inputs z -scored so bar length is effect size per standard deviation; positive favours a CT win), coloured by pillar. The firepower features carry the largest in-sample coefficients, which superficially suggests a dominant skill effect; Sect. 7.8 and the out-of-time holdout show these are the count-confounded features whose apparent importance does not transfer. The economy, defuse-race, and map-control features below them are the ones that generalise.

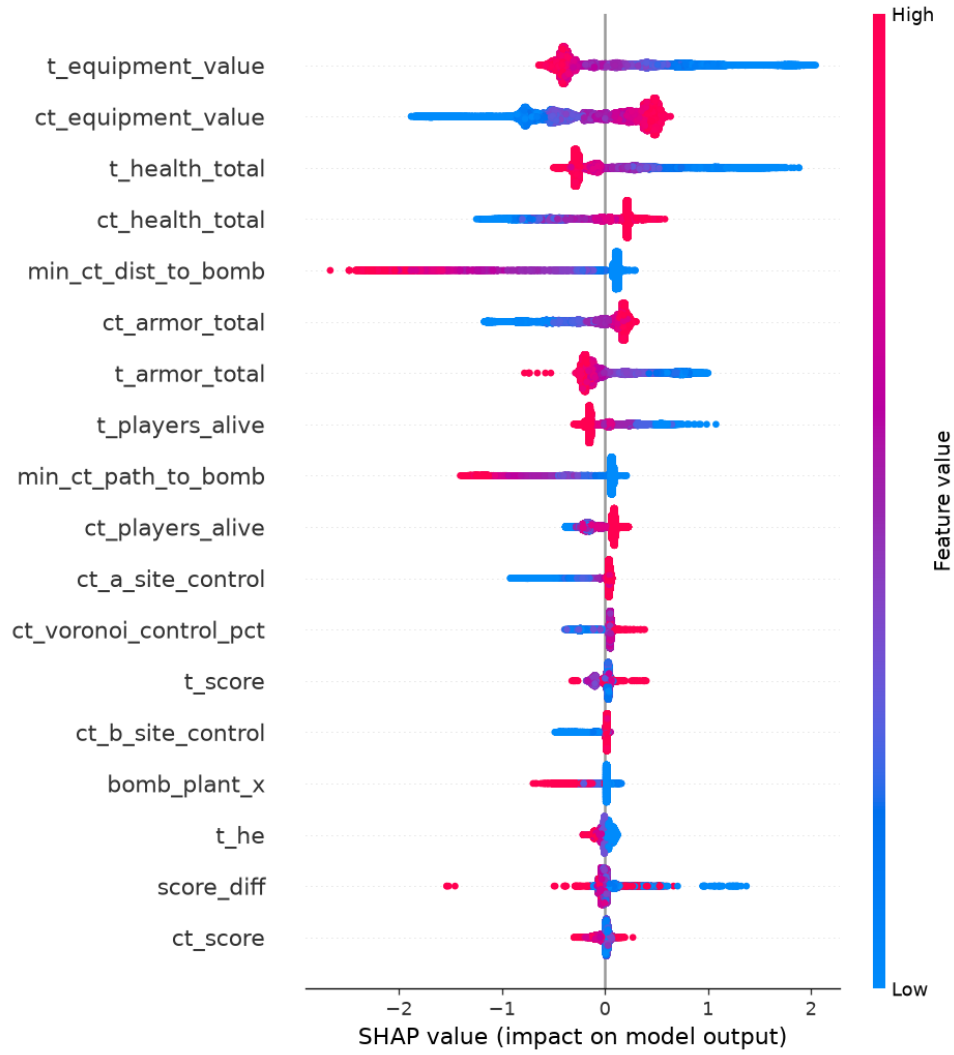


Figure 24: Global SHAP attribution for the gradient-boosted model (set ET). Each point is one snapshot; horizontal position is the feature’s SHAP contribution to $P(\text{CT win})$ and colour is the feature value. The ordering agrees with the permutation-importance analysis: the economy and combat-state variables dominate, with the defuse-race margin the strongest bomb-geometry contributor.